

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\401-V-32003 - 7%.pvdb

Date of Analysis : Aug 11, 2026 2:49pm

PV Elite 28 U2, April 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2040.000 x 2000.000 mm. ID
 Vessel Design Length, Tangent to Tangent 4300.00 mm.
 Distance of Bottom Tangent above Grade 2300.00 mm.
 Distance of Base above Grade 300.00 mm.
 Specified Datum Line Distance 2050.00 mm.
 Internal Design Temperature 80 °C
 Internal Design Pressure 23.000 bars
 External Design Temperature 80 °C
 External Design Pressure 1.013 bars
 Maximum Allowable Working Pressure 24 bars
 External Max. Allowable Working Pressure 6 bars
 Shop Test Pressure 31.353 bars
 Required Minimum Design Metal Temperature -33.0 °C
 Warmest Computed Minimum Design Metal Temperature -33.0 °C
 Wind Design Code IS 875 (Part 3):2015 Amd. 1&2 2020
 Earthquake Design Code Response Spectrum

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	+IT
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-516 70	...	...	K02700	+IT
Nozzle	SA-333 6	...	...	K03006	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Nozzle Flg	SA-333 6	...	...	K03006	+IT
Basering	SA-285 C	...	...	K02801	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
SKIRT	...	...	...	1.5000	N/A	No
Bottom D end	23.000	1.01	24.000	1.5000	No	No
Shell 1	23.000	1.01	25.000	1.5000	N/A	No
shell 2	23.000	1.01	25.000	1.5000	N/A	No
Top D end	23.000	1.01	24.000	1.5000	No	No

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Skirt	-50.0	2000.0	8.0	8.0	...	...	0.70	0.55
Ellipse	0.0	50.0	22.0	19.0	18.2	6.7	1.00	1.00



Cylinder	2800.0	2800.0	20.0	20.0	18.4	10.6	1.00	1.00
Cylinder	4200.0	1400.0	20.0	20.0	18.4	10.6	1.00	1.00
Ellipse	4250.0	50.0	22.0	19.0	18.2	6.7	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3_2"	WRC-107/537	0.858	n/a	Passed
N1A_12"	WRC-297	0.906	Shell	Passed
N1B_12"	WRC-297	0.906	Shell	Passed
N2_18"	WRC-107/537	0.858	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	5457.05	2160.91	25464.3	11288
20	30	2025	4705.01	2108.86	15302	6996.94
30	40	3450	4684	1890.62	15067.3	6892.39
40	50	5550	2940.75	1129.31	4392.45	1602.33
50	60	6275	2352.08	425.339	687.393	21.2674

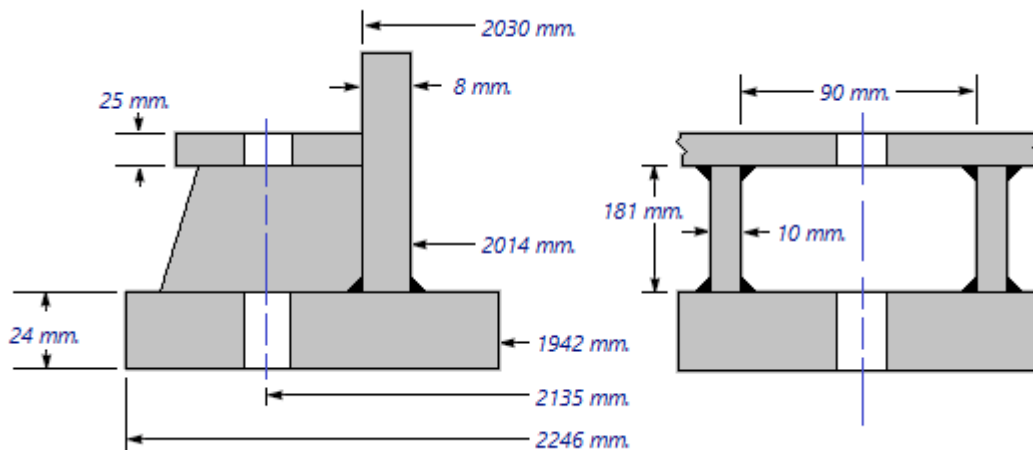
Abs Max of the all of the Stress Ratio's : 0.4707

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing 24.0000 mm.
 Inside Diameter of Basing 1942.0000 mm.
 Outside Diameter of Basing 2246.0000 mm.
 Nominal Diameter of Bolts 30.0000 mm.
 Diameter of Bolt Circle 2135.0000 mm.
 Number of Bolts 16

Thickness of Gusset Plates 10.0000 mm.
 Average Width of Gusset Plates 102.0000 mm.
 Height of Gussets 181.0000 mm.
 Distance between Gussets 90.0000 mm.
 Thickness of Top Plate or Ring 25.0000 mm.
 Radial Width of the Top Plate 102.0000 mm.
 Number of Gussets 32

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View



Loads for Foundation/Support Design:

Total Wind Shear on Support	5457. Kg
Total Earthquake Shear on Support	2161. Kg
Wind Moment on Support	25968. Kg-m.
Earthquake Moment on Support	11792. Kg-m.
Maximum Initial Bolt Load	1924. Kg

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	10963.4 kg.
Shop Test - Fabricated + water (full)	26559.9 kg.
Shipping - Fab. + rem. intls.+ shipping app.	17871.0 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	17871.0 kg.
Empty - Fab. + intls. + details + wghts.	17871.0 kg.
Operating - Empty + operating liquid (no ca)	18071.3 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3_2"	24	...	...	OK	6.30	OK	No Calc[*]
N1A_12"	25	OK	...	OK	9.83	OK	Passed
N1B_12"	25	OK	...	OK	9.83	OK	Passed
MH1_24"	25	OK	...	...		OK	Passed
K1B_1.5"	25	...	...	OK	6.02	OK	No Calc[*]
K1A_1.5"	25	...	...	OK	6.02	OK	No Calc[*]
K2A_3"	25	OK	...	OK	6.30	OK	Passed
K2B_2"	25	OK	...	OK	6.30	OK	Passed
K2C_3"	25	OK	...	...		OK	Passed
K2D_3"	25	OK	...	OK	6.30	OK	Passed
K2E_3"	25	OK	...	OK	6.30	OK	Passed
K2F_3"	25	OK	...	OK	6.30	OK	Passed
K3A_3"	25	OK	...	OK	6.30	OK	Passed
K3B_2"	25	OK	...	OK	6.30	OK	Passed
N4_2"	25	...	...	OK	6.30	OK	No Calc[*]
N2_18"	24	OK	...	OK	9.83	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 24 Nozzle : N218"
 Minimum MAWP Shells/Flanges : 24 Element : Top D end
 Minimum MAPnc Shells/Flanges : 26 Element : Top D end

Computed Vessel M.A.W.P. : 24 bars

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
20	N3_2"	0.000	0.000	119.328
30	N1A_12"	2050.000	180.000	615.600
30	N1B_12"	2050.000	0.000	615.600
30	MH1_24"	800.000	0.000	1225.200
30	K1B_1.5"	200.000	180.000	107.104
30	K1A_1.5"	800.000	180.000	107.104
30	K2A_3"	800.000	90.000	157.130
30	K2B_2"	200.000	90.000	157.130
30	K2C_3"	800.000	270.000	157.130
30	K2D_3"	200.000	270.000	157.130
30	K2E_3"	800.000	110.000	157.130
30	K2F_3"	200.000	110.000	157.130
30	K3A_3"	800.000	236.664	157.130
30	K3B_2"	200.000	236.664	157.130
30	N4_2"	250.000	45.000	121.328
50	N2_18"	0.000	0.000	920.400

The nozzle spacing is computed by the following:

$$= \sqrt{II^2 + Ic^2} \text{ where}$$

II - Arc length along the inside vessel surface in the long. direction.
 Ic - Arc length along the inside vessel surface in the circ. direction

PV Elite 28 U2 Licensee: SL User
FileName : 401-V-32003 - 7%
Nozzle Summary:

Step: 42 2:49pm Aug 11, 2026

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HEXAGON

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Thk mm	Cut Length mm	Flg Class
K1B_1.5"	1.500 in	Actual	LWN	2.760	16.002	...	...	228.58	300
K1A_1.5"	1.500 in	Actual	LWN	2.760	16.002	...	...	228.58	300
N3_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	225.86	300
N4_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	330.32	300
K2A_3"	3.000 in	80	LWN	3.500	7.620	152.40	20.00	324.06	300
K2B_2"	3.000 in	80	LWN	3.500	7.620	152.40	20.00	324.06	300
K2C_3"	3.000 in	80	LWN	3.500	7.620	152.40	20.00	324.06	300
K2D_3"	3.000 in	80	LWN	3.500	7.620	152.40	20.00	324.06	300
K2E_3"	3.000 in	80	LWN	3.500	7.620	152.40	20.00	324.06	300
K2F_3"	3.000 in	80	LWN	3.500	7.620	152.40	20.00	324.06	300
K3A_3"	3.000 in	80	LWN	3.500	7.620	152.40	20.00	377.29	300
K3B_2"	3.000 in	80	LWN	3.500	7.620	152.40	20.00	377.29	300
N1A_12"	12.000 in	Actual	WNF	12.945	12.000	560.00	20.00	235.08	300
N1B_12"	12.000 in	Actual	WNF	12.945	12.000	560.00	20.00	235.08	300
N2_18"	18.000 in	Actual	WNF	18.945	12.000	800.00	19.00	457.79	300
MH1_24"	24.000 in	Actual	WNF	24.945	12.000	1140.00	20.00	284.88	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1B_1.5"	SA-350 LF2	20.000	10.000	...	...	...
K1A_1.5"	SA-350 LF2	20.000	10.000	...	...	...
N3_2"	SA-350 LF2	19.000	10.000	...	...	...
N4_2"	SA-333 6	20.000	10.000	...	...	...
K2A_3"	SA-333 6	20.000	10.000	16.000	20.000	...
K2B_2"	SA-333 6	20.000	10.000	16.000	20.000	...
K2C_3"	SA-333 6	20.000	10.000	16.000	20.000	...
K2D_3"	SA-333 6	20.000	10.000	16.000	20.000	...
K2E_3"	SA-333 6	20.000	10.000	16.000	20.000	...
K2F_3"	SA-333 6	20.000	10.000	16.000	20.000	...
K3A_3"	SA-333 6	20.000	10.000	16.000	20.000	...
K3B_2"	SA-333 6	20.000	10.000	16.000	20.000	...
N1A_12"	SA-516 70	20.000	10.000	16.000	20.000	...
N1B_12"	SA-516 70	20.000	10.000	16.000	20.000	...
N2_18"	SA-516 70	19.000	10.000	16.000	19.000	...
MH1_24"	SA-516 70	20.000	10.000	16.000	20.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
K1B_1.5"	150.000	180.0	207.96	0.00	Shell 1
K1A_1.5"	750.000	180.0	207.96	0.00	Shell 1
N3_2"	...	0.0	206.38	0.00	Bottom D end
N4_2"	200.000	45.0	309.43	0.00	Shell 1
K2A_3"	750.000	90.0	303.08	0.00	Shell 1



K2B_2"	150.000	90.0	303.08	0.00	Shell 1
K2C_3"	750.000	270.0	303.08	0.00	Shell 1
K2D_3"	150.000	270.0	303.08	0.00	Shell 1
K2E_3"	750.000	110.0	303.08	0.00	Shell 1
K2F_3"	150.000	110.0	303.08	0.00	Shell 1
K3A_3"	750.000	236.7	303.08	0.00	Shell 1
K3B_2"	150.000	236.7	303.08	0.00	Shell 1
N1A_12"	2000.000	180.0	201.48	0.00	Shell 1
N1B_12"	2000.000	0.0	201.48	0.00	Shell 1
N2_18"	...	0.0	422.77	0.00	Top D end
MH1_24"	750.000	0.0	213.38	0.00	Shell 1



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 22.0mm. THK X 2000.0mm. ID X 50.0mm.	SA-516 70
1	CYLINDER: 20.0mm. THK X 2000.0mm. ID X 1400.0mm.	SA-516 70
2	INSULATION: 550mm X 115mm THK	CC
1	PLATFORM: 1200mm X 1500mm WIDE	
1	INSULATION: 2800mm X 115mm THK	CC
1	INSULATION: 1400mm X 115mm THK	CC
1	PLATFORM: 1200mm X 4038mm WIDE	
1	LADDER: 7200.0mm	...
1	BASERING: 2246.0mm. OD X 1942.0 ID X 24.0mm.	SA-285 C
16	BASERING BOLTS: 30.0mm. DIA	SA-36
16	BASERING BOLT NUTS: 30.0mm. DIA	...
1	BASERING TOP RING: 2234.0mm. OD X 2030.0 ID X 25.0mm.	...
32	BASERING GUSSET PLATES: 102.0mm. X 102.0mm. X 181.0mm. X 10.0mm.	...
2	CYLINDRICAL SKIRT SEGMENT 8.0mm. THK X 3069.3mm. X 2000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 8.0mm. THK X 188.5mm. X 2000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm. THK X 2967.0mm. X 2800.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm. THK X 3316.1mm. X 2800.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Notes								
Bottom D end [10]	!	-33	-33		1.000	19.000	1.00	No
Bottom D end [7]	!	-33	-41		0.863	22.000	1.00	No
Shell 1 [8]	!	-33	-35		0.957	20.000	1.00	No
shell 2 [8]	!	-33	-35		0.956	20.000	1.00	No
Top D end [10]	!	-33	-33		1.000	19.000	1.00	No
Top D end [7]	!	-33	-41		0.863	22.000	1.00	No
N32" [1]	!	-46	-104		0.030	16.764	1.00	No
Nozzle Flg [5]	!	-46	-46					
N1A12" [1]	!	-33	-38		0.911	20.000	1.00	No
Nozzle Flg [5]	!	-46	-85					
N1B12" [1]	!	-33	-38		0.911	20.000	1.00	No
Nozzle Flg [5]	!	-46	-85					
MH124" [1]	!	-33	-38		0.911	20.000	1.00	No
Nozzle Flg [5]	!	-46	-85					
K1B1.5" [1]	!	-46	-104		0.024	16.002	1.00	No
Nozzle Flg [5]	!	-46	-46					
K1A1.5" [1]	!	-46	-104		0.024	16.002	1.00	No
Nozzle Flg [5]	!	-46	-46					
K2A3" [1]	!	-33	-38		0.911	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K2B2" [1]	!	-33	-38		0.911	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K2C3" [1]	!	-33	-38		0.911	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K2D3" [1]	!	-33	-38		0.911	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K2E3" [1]	!	-33	-38		0.911	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K2F3" [1]	!	-33	-38		0.911	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K3A3" [1]	!	-33	-38		0.911	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
K3B2" [1]	!	-33	-38		0.911	20.000	1.00	No
Nozzle Flg [5]	!	-46	-46					
N42" [1]	!	-46	-104		0.035	16.764	1.00	No
Nozzle Flg [5]	!	-46	-46					
N218" [1]	!	-33	-41		0.859	19.000	1.00	No
Nozzle Flg [5]	!	-46	-85					
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No

Warmest MDMT: -33 -33

Required Minimum Design Metal Temperature -33.0 °C
 Warmest Computed Minimum Design Metal Temperature -33.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)



[20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
[21] - Body Flange Bolting Material
[22] - Nozzle Flange Bolting Material
[23] - Stiffening Ring to Shell Weld
[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 10 30 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest) 23 bars
 Design Internal Temperature 80.0 °C
 Projection of Nozzle from Vessel Top 0 mm.
 Projection of Nozzle from Vessel Bottom 0 mm.
 Minimum Design Metal Temperature -33.0 °C
 Type of Construction Welded
 Special Service None
 Degree of Radiography RT-1
 Use Higher Longitudinal Stresses (Flag) Y
 Select t for Internal Pressure (Flag) N
 Select t for External Pressure (Flag) N
 Select t for Axial Stress (Flag) N
 Select Location for Stiff. Rings (Flag) N
 Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Load Case 1 NP+EW+1.1WI+FW+BW
 Load Case 2 NP+EW+EE+FS+BS
 Load Case 3 NP+OW+1.1WI+FW+BW
 Load Case 4 NP+OW+EQ+FS+BS
 Load Case 5 NP+HW+HI
 Load Case 6 NP+HW+HE
 Load Case 7 IP+OW+1.1WI+FW+BW
 Load Case 8 IP+OW+EQ+FS+BS
 Load Case 9 EP+OW+1.1WI+FW+BW
 Load Case 10 EP+OW+EQ+FS+BS
 Load Case 11 HP+HW+HI
 Load Case 12 HP+HW+HE
 Load Case 13 IP+WE+EW
 Load Case 14 IP+WF+CW
 Load Case 15 IP+VO+OW
 Load Case 16 IP+VE+EW
 Load Case 17 NP+VO+OW
 Load Case 18 FS+BS+IP+OW
 Load Case 19 FS+BS+EP+OW
 Load Case 20 BL+IP+OW



Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm.
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Seismic Design Code		Spectrum
Range Type		Period
Ordinate Type		Acceleration
Include Missing Mass Components		Yes
Shock Scale Factor X Direction		1.0000
Shock Scale Factor Y Direction		0.6700
Combination Method		Group
Spectrum Name		User Defined

User Defined Spectrum Data Points:

Acceleration	Period
0.0700	0.0100
0.1170	0.0500
0.1750	0.1000
0.1750	0.1300
0.1750	0.2000
0.1750	0.2700
0.1750	0.3000
0.1750	0.4000
0.1400	0.5000
0.1270	0.5500
0.1170	0.6000
0.1050	0.6700
0.0870	0.8000
0.0700	1.0000
0.0580	1.2000
0.0500	1.4000
0.0440	1.6000
0.0390	1.8000
0.0350	2.0000
0.0290	2.4000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes



Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	SKIRT
Distance "FROM" to "TO"	2000 mm.
Skirt Outside Diameter	2030 mm.
Diameter of Skirt at Base	2030 mm.
Skirt Thickness	8 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	8 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	80 °C
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm2
Allowable Stress, Operating	138 N./mm2
Allowable Stress, Hydrotest	179.4 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	243 N./mm2
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.55
Weld is pre-Heated	No
Element From Node	10
Detail Type	Weight
Detail ID	skirt Opening
Dist. from "FROM" Node / Offset dist	1000 mm.
Miscellaneous Weight	200 Kgf
Offset from Element Centerline	0 mm.

Element#	2/5
Element From Node	20
Element To Node	30
Element Type	Elliptical
Description	Bottom D end
Distance "FROM" to "TO"	50 mm.
Inside Diameter	2000 mm.
Element Thickness	19 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	22 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	23 bars
Design Temperature Internal Pressure	80 °C
Design External Pressure	1.0132 bars
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Insulation
Detail ID	Insul.0



Dist. from "FROM" Node / Offset dist	-500	mm.
Height/Length of Insulation	550	mm.
Thickness of Insulation	115	mm.
Density	0.40000E-04	kg./cm3

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N32"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	47.877	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	

Element From Node	20	
Detail Type	Weight	
Detail ID	vortex breaker	
Dist. from "FROM" Node / Offset dist	-450	mm.
Miscellaneous Weight	0.3	Kgf
Offset from Element Centerline	0	mm.

Element#	3/5	
Element From Node	30	
Element To Node	40	
Element Type	Cylinder	
Description	Shell 1	
Distance "FROM" to "TO"	2800	mm.
Element Outside Diameter	2040	mm.
Element Thickness	20	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	20	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	23	bars
Design Temperature Internal Pressure	80	°C
Design External Pressure	1.0132	bars
Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Platform	
Detail ID	PLAT 1	
Dist. from "FROM" Node / Offset dist	0	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	43200	cm2
Platform Weight	3238.1	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm2
Platform Width	1500	mm.
Platform Height	1200	mm.
Platform Clearance or End Offset	230	mm.
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	0	mm.
Unit Weight of Ladder	0	Kgf/mm.
Platform Length (top head platform)	0	mm.

Element From Node	30	
Detail Type	Insulation	



Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	2800	mm.
Thickness of Insulation	115	mm.
Density	0.40000E-04	kg./cm3

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1A12"	
Dist. from "FROM" Node / Offset dist	2000	mm.
Nozzle Diameter	12	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	113	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+IT

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1B12"	
Dist. from "FROM" Node / Offset dist	2000	mm.
Nozzle Diameter	12	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	113	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+IT

Element From Node	30	
Detail Type	Nozzle	
Detail ID	MH124"	
Dist. from "FROM" Node / Offset dist	750	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	742.66	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+IT

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K1B1.5"	
Dist. from "FROM" Node / Offset dist	150	mm.
Nozzle Diameter	1.5	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	8.3591	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K1A1.5"	
Dist. from "FROM" Node / Offset dist	750	mm.
Nozzle Diameter	1.5	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	8.3591	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT



Element From Node 30
Detail Type Nozzle
Detail ID K2A3"
Dist. from "FROM" Node / Offset dist 750 mm.
Nozzle Diameter 3 in.
Nozzle Schedule 80
Nozzle Class 300
Layout Angle 90.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 14.285 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-333 6 +IT

Element From Node 30
Detail Type Nozzle
Detail ID K2B2"
Dist. from "FROM" Node / Offset dist 150 mm.
Nozzle Diameter 3 in.
Nozzle Schedule 80
Nozzle Class 300
Layout Angle 90.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 14.285 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-333 6 +IT

Element From Node 30
Detail Type Nozzle
Detail ID K2C3"
Dist. from "FROM" Node / Offset dist 750 mm.
Nozzle Diameter 3 in.
Nozzle Schedule 80
Nozzle Class 300
Layout Angle 270.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 14.285 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-333 6 +IT

Element From Node 30
Detail Type Nozzle
Detail ID K2D3"
Dist. from "FROM" Node / Offset dist 150 mm.
Nozzle Diameter 3 in.
Nozzle Schedule 80
Nozzle Class 300
Layout Angle 270.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 14.285 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-333 6 +IT

Element From Node 30
Detail Type Nozzle
Detail ID K2E3"
Dist. from "FROM" Node / Offset dist 750 mm.
Nozzle Diameter 3 in.
Nozzle Schedule 80
Nozzle Class 300
Layout Angle 110.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 14.285 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-333 6 +IT

Element From Node 30
Detail Type Nozzle
Detail ID K2F3"
Dist. from "FROM" Node / Offset dist 150 mm.



Nozzle Diameter 3 in.
 Nozzle Schedule 80
 Nozzle Class 300
 Layout Angle 110.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 14.285 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-333 6 +IT

Element From Node 30
 Detail Type Nozzle
 Detail ID K3A3"
 Dist. from "FROM" Node / Offset dist 750 mm.
 Nozzle Diameter 3 in.
 Nozzle Schedule 80
 Nozzle Class 300
 Layout Angle 236.664
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 14.285 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-333 6 +IT

Element From Node 30
 Detail Type Nozzle
 Detail ID K3B2"
 Dist. from "FROM" Node / Offset dist 150 mm.
 Nozzle Diameter 3 in.
 Nozzle Schedule 80
 Nozzle Class 300
 Layout Angle 236.664
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 14.285 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-333 6 +IT

Element From Node 30
 Detail Type Nozzle
 Detail ID N42"
 Dist. from "FROM" Node / Offset dist 200 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 45.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-333 6 +IT

Element From Node 30
 Detail Type Weight
 Detail ID DeflectorN1B
 Dist. from "FROM" Node / Offset dist 2100 mm.
 Miscellaneous Weight 15 Kgf
 Offset from Element Centerline 850 mm.

Element From Node 30
 Detail Type Weight
 Detail ID DeflectorN1A
 Dist. from "FROM" Node / Offset dist 2100 mm.
 Miscellaneous Weight 15 Kgf
 Offset from Element Centerline 850 mm.

Element From Node 30
 Detail Type Weight
 Detail ID Manway Davit
 Dist. from "FROM" Node / Offset dist 750 mm.
 Miscellaneous Weight 360 Kgf
 Offset from Element Centerline 1400 mm.

Element From Node 30



Detail Type	Weight
Detail ID	Ladder rung
Dist. from "FROM" Node / Offset dist	2100 mm.
Miscellaneous Weight	12 Kg
Offset from Element Centerline	0 mm.

Element#	4/5
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	shell 2
Distance "FROM" to "TO"	1400 mm.
Inside Diameter	2000 mm.
Element Thickness	20 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	20 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	23 bars
Design Temperature Internal Pressure	80 °C
Design External Pressure	1.0132 bars
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	1400 mm.
Thickness of Insulation	115 mm.
Density	0.40000E-04 kg./cm ³

Element From Node	40
Detail Type	Weight
Detail ID	demister
Dist. from "FROM" Node / Offset dist	1100 mm.
Miscellaneous Weight	200 Kg
Offset from Element Centerline	0 mm.

Element From Node	40
Detail Type	Weight
Detail ID	clit Wt
Dist. from "FROM" Node / Offset dist	1100 mm.
Miscellaneous Weight	1000 Kg
Offset from Element Centerline	0 mm.

Element From Node	40
Detail Type	Weight
Detail ID	Lifting Wt
Dist. from "FROM" Node / Offset dist	1200 mm.
Miscellaneous Weight	300 Kg
Offset from Element Centerline	0 mm.

Element#	5/5
Element From Node	50
Element To Node	60
Element Type	Elliptical
Description	Top D end
Distance "FROM" to "TO"	50 mm.
Inside Diameter	2000 mm.
Element Thickness	19 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	22 mm.



External Corrosion Allowance	0	mm.
Design Internal Pressure	23	bars
Design Temperature Internal Pressure	80	°C
Design External Pressure	1.0132	bars
Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	50	
Detail Type	Platform	
Detail ID	PLAT 1	
Dist. from "FROM" Node / Offset dist	900	mm.
Platform Start Angle (degrees)	45.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	93226	cm2
Platform Weight	3039.8	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm2
Platform Width	4038	mm.
Platform Height	1200	mm.
Platform Clearance or End Offset	2000	mm.
Platform Force Coefficient	1.2	
Ladder Layout Angle	90.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	7200	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	4038	mm.
Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	550	mm.
Thickness of Insulation	115	mm.
Density	0.40000E-04	kg./cm3
Element From Node	50	
Detail Type	Nozzle	
Detail ID	N218"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	18	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	247.25	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+IT



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
	SKIRT	...	2000	...	2000
	Bottom D end	...	2050	...	50
	Shell 1	...	4850	...	2800
	shell 2	...	6250	...	1400
	Top D end	...	6300	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
SKIRT		...	8	1.5	2030	...	1
Bottom D end		23	22	1.5	2000	138	2
Shell 1		23	20	1.5	2040	138	3
shell 2		23	20	1.5	2000	138	4
Top D end		23	22	1.5	2000	138	5

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
SKIRT		...	No Calc	No Calc	8	No Calc	1
Bottom D end		23	24.1178	26.1682	19	18.1876	2
Shell 1		23	25.2103	27.2705	20	18.3888	3
shell 2		23	25.2103	27.2705	20	18.3616	4
Top D end		23	24.1178	26.1682	19	18.1876	5

Minimum 24 26

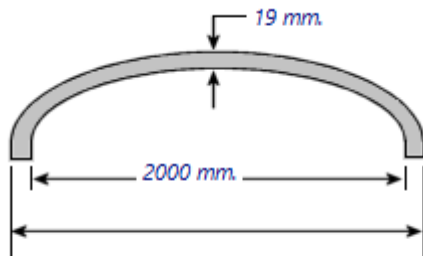
MAWP: 24 bars, limited by: Top D end.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom D end

Elliptical Head Design Information:					
Design Pressure	P	23.000 bars	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2000.000 mm.	Actual Stress		131.6 N./mm^2
Spec. Min. Thk t _e , or sa		19.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		48498.9 cm^2	SG of Contents		0.000
Element Volume		1204493.0 cm3	Weight of Contents		0.0 Kg
Empty Weight		888.5 Kg	Operating Weight		888.5 Kg
			Comp. All. Ope. Stress		126.4 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 25.000 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 12.500 mm.

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (23.0 \cdot 2003.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 23.0)$$

$$= 16.6876 + 1.5000 = 18.1876 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 17.5) / (0.998 \cdot 2003.0 + 0.2 \cdot 17.5)$$

$$= 24 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 19.0) / (1.0 \cdot 2000.0 + 0.2 \cdot 19.0)$$

$$= 26 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (23.0 \cdot (0.998 \cdot 2003.0 + 0.2 \cdot 17.5)) / (2 \cdot 1.0 \cdot 17.5)$$

$$= 131.601 \text{ N./mm}^2$$



Straight Flange Required Thickness:
= $(P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
= $(23.0 \cdot 1001.5) / (138.0 \cdot 1.0 - 0.6 \cdot 23.0) + 1.5$
= 18.362 mm.

Straight Flange Maximum Allowable Working Pressure:
= $(S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
= $(138.0 \cdot 1.0 \cdot 20.5) / (1001.5 + 0.6 \cdot 20.5)$
= 28 bars

Factor K, corroded condition [Kcor]:
= $(2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
= $(2 + (2003.0 / (2 \cdot 501.5))^2) / 6$
= 0.998007

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.701 %

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 25.000 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 12.500 mm.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Bottom D end

Elliptical Head From 20 to 30 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
17.500	2038.00	116.46	0.0011926	90.53

$$MAEP = B / (K0 \cdot Do / t) = 90.5313 / (0.9 \cdot 116.4571) = 8.637 \text{ bars}$$

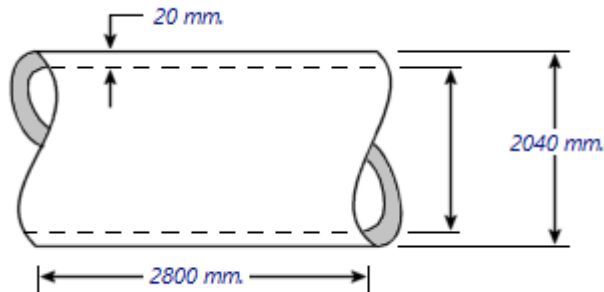
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
5.223	2038.00	390.20	0.0003559	35.59

$$MAEP = B / (K0 \cdot Do / t) = 35.5863 / (0.9 \cdot 390.2001) = 1.0133 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 1

Cylindrical Shell Design Information:					
Design Pressure	P	23.000 bars	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		2800.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Outside Diameter	Do	2040.000 mm.	Actual Stress		125.9 N./mm^2
Spec. Min. Thk t _e , or s _a		20.0000 mm.	Spec. Nominal Thk.		20.000 mm.
Surface Area		179447.8 cm^2	SG of Contents		0.000
Element Volume		8798036.0 cm3	Weight of Contents		0.0 Kgf
Empty Weight		2754.4 Kgf	Operating Weight		2754.4 Kgf
			Comp. All. Ope. Stress		127.8 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (23.0 \cdot 1020.0) / (138.0 \cdot 1.0 + 0.4 \cdot 23.0)$$

$$= 16.8888 + 1.5000 = 18.3888 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.5) / (1020.0 - 0.4 \cdot 18.5)$$

$$= 25 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.0) / (1020.0 - 0.4 \cdot 20.0)$$

$$= 27 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

$$= (23.0 \cdot (1020.0 - 0.4 \cdot 18.5)) / (1.0 \cdot 18.5)$$

$$= 125.898 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 0.990 %

Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell 1

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.500	2040.00	4633.33	110.27	2.2712	0.0004889	48.88

$$MAEP = (4*B)/(3*(Do/t)) = (4*48.879)/(3*110.2703) = 5.9099 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.115	2040.00	4633.33	223.80	2.2712	0.0001701	17.01

$$MAEP = (4*B)/(3*(Do/t)) = (4*17.0082)/(3*223.7997) = 1.0132 \text{ bars}$$

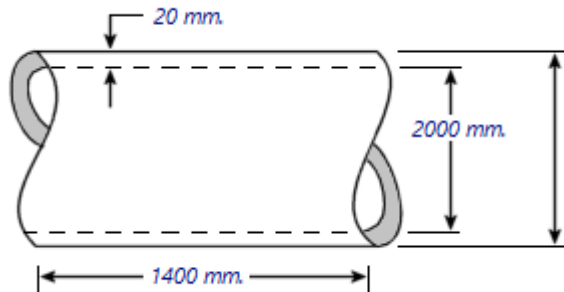
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.500	2040.00	96491.90	110.27	47.2999	0.0000921	9.21

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.2077)/(3*110.2703) = 1.1133 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell 2

Cylindrical Shell Design Information:					
Design Pressure	P	23.000 bars	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		1400.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2000.000 mm.	Actual Stress		125.9 N./mm^2
Spec. Min. Thk t _e , or sa		20.0000 mm.	Spec. Nominal Thk.		20.000 mm.
Surface Area		89723.9 cm^2	SG of Contents		0.000
Element Volume		4399018.0 cm3	Weight of Contents		0.0 Kgf
Empty Weight		1377.2 Kgf	Operating Weight		1377.2 Kgf
			Comp. All. Ope. Stress		127.8 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (23.0 \cdot 1001.5) / (138.0 \cdot 1.0 - 0.6 \cdot 23.0)$$

$$= 16.8616 + 1.5000 = 18.3616 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.5) / (1001.5 + 0.6 \cdot 18.5)$$

$$= 25 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.0) / (1000.0 + 0.6 \cdot 20.0)$$

$$= 27 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (23.0 \cdot (1001.5 + 0.6 \cdot 18.5)) / (1.0 \cdot 18.5)$$

$$= 125.898 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.990 %

Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: shell 2

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.500	2040.00	4633.33	110.27	2.2712	0.0004889	48.88

$$MAEP = (4*B)/(3*(Do/t)) = (4*48.879)/(3*110.2703) = 5.9099 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.115	2040.00	4633.33	223.80	2.2712	0.0001701	17.01

$$MAEP = (4*B)/(3*(Do/t)) = (4*17.0082)/(3*223.7997) = 1.0132 \text{ bars}$$

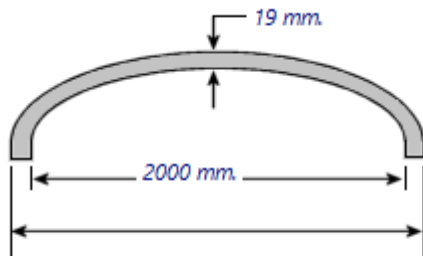
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.500	2040.00	96491.90	110.27	47.2999	0.0000921	9.21

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.2077)/(3*110.2703) = 1.1133 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top D end

Elliptical Head Design Information:					
Design Pressure	P	23.000 bars	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2000.000 mm.	Actual Stress		131.6 N./mm^2
Spec. Min. Thk t _e , or sa		19.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		48498.9 cm^2	SG of Contents		0.000
Element Volume		1204493.0 cm3	Weight of Contents		0.0 Kgf
Empty Weight		888.5 Kgf	Operating Weight		888.5 Kgf
			Comp. All. Ope. Stress		126.4 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 25.000 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 12.500 mm.

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (23.0 \cdot 2003.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 23.0)$$

$$= 16.6876 + 1.5000 = 18.1876 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 17.5) / (0.998 \cdot 2003.0 + 0.2 \cdot 17.5)$$

$$= 24 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 19.0) / (1.0 \cdot 2000.0 + 0.2 \cdot 19.0)$$

$$= 26 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (23.0 \cdot (0.998 \cdot 2003.0 + 0.2 \cdot 17.5)) / (2 \cdot 1.0 \cdot 17.5)$$

$$= 131.601 \text{ N./mm}^2$$



Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (23.0 \cdot 1001.5) / (138.0 \cdot 1.0 - 0.6 \cdot 23.0) + 1.5$
 $= 18.362 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138.0 \cdot 1.0 \cdot 20.5) / (1001.5 + 0.6 \cdot 20.5)$
 $= 28 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (2003.0 / (2 \cdot 501.5))^2) / 6$
 $= 0.998007$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.701 %

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 25.000 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 12.500 mm.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Top D end

Elliptical Head From 50 to 60 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
17.500	2038.00	116.46	0.0011926	90.53

$$\text{MAEP} = B / (K0 \cdot Do / t) = 90.5313 / (0.9 \cdot 116.4571) = 8.637 \text{ bars}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
5.223	2038.00	390.20	0.0003559	35.59

$$\text{MAEP} = B / (K0 \cdot Do / t) = 35.5863 / (0.9 \cdot 390.2001) = 1.0133 \text{ bars}$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	31.353 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	29.900 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	33.823 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	26.530 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	32.890 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	32.890 bars
App 27-4	= M.A.W.P.	24.118 bars

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 24.118 • 1.0
 = 31.353 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom D end	166.4	179.4	0.927	31.55
Shell 1	159.7	179.4	0.890	31.55
shell 2	159.6	179.4	0.890	31.55
Top D end	166.4	179.4	0.927	31.55

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_2"	Nozzle	138.00	138.00	1.000
N1A_12"	Nozzle	138.00	138.00	1.000
N1A_12"	Pad	138.00	138.00	1.000
N1B_12"	Nozzle	138.00	138.00	1.000
N1B_12"	Pad	138.00	138.00	1.000
MH1_24"	Nozzle	138.00	138.00	1.000
MH1_24"	Pad	138.00	138.00	1.000
K1B_1.5"	Nozzle	138.00	138.00	1.000
K1A_1.5"	Nozzle	138.00	138.00	1.000
K2A_3"	Nozzle	118.00	118.00	1.000
K2A_3"	Pad	138.00	138.00	1.000
K2B_2"	Nozzle	118.00	118.00	1.000
K2B_2"	Pad	138.00	138.00	1.000
K2C_3"	Nozzle	118.00	118.00	1.000
K2C_3"	Pad	138.00	138.00	1.000
K2D_3"	Nozzle	118.00	118.00	1.000
K2D_3"	Pad	138.00	138.00	1.000
K2E_3"	Nozzle	118.00	118.00	1.000
K2E_3"	Pad	138.00	138.00	1.000
K2F_3"	Nozzle	118.00	118.00	1.000
K2F_3"	Pad	138.00	138.00	1.000
K3A_3"	Nozzle	118.00	118.00	1.000
K3A_3"	Pad	138.00	138.00	1.000
K3B_2"	Nozzle	118.00	118.00	1.000
K3B_2"	Pad	138.00	138.00	1.000
N4_2"	Nozzle	118.00	118.00	1.000
N2_18"	Nozzle	138.00	138.00	1.000
N2_18"	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
-------------	---------	-----------	-------



Bottom D end	138.00	138.00	1.000
Shell 1	138.00	138.00	1.000
shell 2	138.00	138.00	1.000
Top D end	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm2):

Description	Ambient	Operating	Ratio
N3_2"	6.67	179.40	0.037
N1A_12"	41.97	179.40	0.234
N1B_12"	41.97	179.40	0.234
MH1_24"	82.04	179.40	0.457
K1B_1.5"	5.65	179.40	0.031
K1A_1.5"	5.65	179.40	0.031
K2A_3"	19.77	153.40	0.129
K2B_2"	19.77	153.40	0.129
K2C_3"	19.77	153.40	0.129
K2D_3"	19.77	153.40	0.129
K2E_3"	19.77	153.40	0.129
K2F_3"	19.77	153.40	0.129
K3A_3"	19.77	153.40	0.129
K3B_2"	19.77	153.40	0.129
N4_2"	6.67	153.40	0.044
N2_18"	62.00	179.40	0.346

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	2000	...	...	No Calc	No Calc
20	30	No Calc	2038	17.5	0.0011926	90.5313
30	40	4633.33	2040	18.5	0.0004889	48.879
40	50	4633.33	2040	18.5	0.0004889	48.879
50	60	No Calc	2038	17.5	0.0011926	90.5313

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	19	6.72296	1.0132	8.63704
30	40	20	10.6153	1.0132	5.90986
40	50	20	10.6153	1.0132	5.90986
50	60	19	6.72296	1.0132	8.63704

Minimum

6

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	2000	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	4633.33	96491.9	No Calc	No Calc
40	50	4633.33	96491.9	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	1151.9	...	1004.67	...	80.6329
20	30	888.521	1204493	827.94	1209685	62.1965
30	40	2754.36	8798036	2549.68	8824449	192.805
40	50	1377.18	4399018	1274.84	4412224	96.4026
50	60	888.521	1204493	827.94	1209685	62.1965
Total		7060	15606039.00	6485	15656045.00	494

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset, Dtl. Cent. mm.	Description
10	Wght	200	...	1000	...	skirt Opening
20	Ins1	29.6318	...	-225	...	Insul.
20	Noz1	51.2289	...	-643.687	...	N3_2"
20	Wght	0.3	...	-450	...	vortex breaker
30	Plat	3464.8	0.18371E-04	...	...	PLAT 1
30	Ins1	87.3014	...	1400	...	Ins: 20
30	Noz1	120.912	-1100.74	2000	...	N1A_12"
30	Noz1	120.912	1100.74	2000	...	N1B_12"
30	Noz1	794.642	1106.69	750	...	MH1_24"
30	Noz1	8.94429	-1103.97	150	...	K1B_1.5"
30	Noz1	8.94429	-1103.97	750	...	K1A_1.5"
30	Noz1	15.2851	...	750	-1151.54	K2A_3"
30	Noz1	15.2851	...	150	-1151.54	K2B_2"
30	Noz1	15.2851	...	750	1151.54	K2C_3"
30	Noz1	15.2851	...	150	1151.54	K2D_3"
30	Noz1	15.2851	-393.849	750	-1082.09	K2E_3"
30	Noz1	15.2851	-393.849	150	-1082.09	K2F_3"
30	Noz1	15.2851	-632.824	750	962.065	K3A_3"
30	Noz1	15.2851	-632.824	150	962.065	K3B_2"
30	Noz1	14.2993	816.504	200	-816.504	N4_2"
30	Wght	15	850	2100	...	Deflector_N1B
30	Wght	15	-850	2100	...	Deflector_N1A
30	Wght	360	1400	750	...	Manway Davit
30	Wght	12	...	2100	...	Ladder rung
40	Ins1	43.6507	...	700	...	Ins: 30
40	Wght	200	...	1100	...	demister
40	Wght	1000	...	1100	...	clit Wt
40	Wght	300	...	1200	...	Lifting Wt
50	Plat	3252.53	19	900	...	PLAT 1
50	Ins1	29.6318	...	275	...	Ins: 40
50	Noz1	264.554	...	751.891	...	N2_18"

Total Weight of Each Detail Type:

Platforms	6717.3
Insulation	190.2
Nozzles	1506.7
Weights	2102.3

Sum of the Detail Weights 10516.6 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	7554.7	7554.7	7554.7	7554.7	7554.7	7554.7
Nozzles	1506.7	1506.7	1506.7	1506.7	1506.7	1506.7
Wld Weights	1902.0	1902.0	1902.0	1902.0	1902.0	1902.0



Platforms	...	...	6717.3	6717.3	6717.3	6717.3
Insulation	...	...	190.2	190.2	190.2	190.2
Ope Weights	...	...	...	...	...	200.3
Test Liquid	...	15596.5	...	...	...	...
Totals	10963.4	26559.9	17871.0	17871.0	17871.0	18071.3

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	10963.4 kg.
Shop Test Wt.	- Fabricated weight + water (full)	26559.9 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	17871.0 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	17871.0 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	17871.0 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	18071.3 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	7993.9 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	14982.1 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	8019.1 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	127549	126543
20	30	48498.9	46501
30	40	179448	175929
40	50	89723.9	87964.6
50	60	48498.9	46501

Total	493718.2	483439.2 cm2
Total	531.4	520.4 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	1432.53	1432.53	1432.53	...	1432.53
20	30	1031.58	1031.88	2235.34	...	1031.88
30	40	7690.2	8092.2	16482.9	1343.99	8092.2
40	50	1517.23	3017.23	5913.57	...	3017.23
50	60	4497.44	4497.44	5701.19	61.7994	4497.44

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	18071.3	18071.3	31765.5
20	30	16638.7	16638.7	30333
30	40	15606.9	15606.9	28097.6
40	50	7514.67	7514.67	11614.8
50	60	4497.44	4497.44	5701.19



The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	1405.79	1405.79	1405.79
20	30	1405.79	1405.79	1405.79
30	40	1405.79	1405.79	1405.79
40	50	61.7994	61.7994	61.7994
50	60	61.7994	61.7994	61.7994

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N3_2"	47.94	51.00	80	300	GR 1.1	...	...	...	...
N1A_12"	47.94	51.00	80	300	GR 1.1	...	...	...	...
N1B_12"	47.9	51.0	80	300	GR 1.1	21.0	47.9	47.9	47.9
MH1_24"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K1B_1.5"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K1A_1.5"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K2A_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K2B_2"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K2C_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K2D_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K2E_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K2F_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K3A_3"	47.94	51.00	80	300	GR 1.1	...	...	...	...
K3B_2"	47.94	51.00	80	300	GR 1.1	...	...	...	...
N4_2"	47.94	51.00	80	300	GR 1.1	...	...	...	...
N2_18"	47.94	51.00	80	300	GR 1.1	...	...	...	...
Min Rating	47.941	51.000, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N1B_12"	357.251	0.50	1839	830	24.1	47.9	Pass

Evaluating Flange on Nozzle [N1B12"]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR) \\
 &= 16(1839) + 4 \cdot \text{abs}(830) \cdot 357.251 \leq \pi \cdot 357.251^3 \cdot ((47.941 - 24.118) + 0.500 \cdot 47.941) \\
 &= 0.30610E+05 \leq 0.69814E+05 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle N1B12" [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(47.941, 0.500 \cdot 47.941 + 47.941 - 30610.0 / (\pi \cdot 357.251^3)) \\
 &= \min(47.941, 50.956) \\
 &= 47.941 \text{ bars}
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 12.414 Hz.

The Natural Frequency for the Vessel (Ope...) is 12.414 Hz.

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (Part 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm.
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Height to Diameter ratio: 3.161

Checking the requirement for dynamic effect of wind (9.1):

Vessel Operating Natural Frequency: 12.414 Hz

Note: Vessel Natural frequency is ≥ 1 Hz.

Note: Vessel height to diameter ratio (3.161) is ≤ 5 .

Note: Dynamic wind analysis is not required.

Design Wind Speed [Vz]:

$$\begin{aligned} &= \text{Basic Wind Speed} \cdot k_1 \cdot k_2 \cdot k_3 \cdot k_4 \quad (\text{for the 1st element}) \\ &= 50.0 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 1.15 \\ &= 57.496 \text{ meters/s} \end{aligned}$$

Design Wind Pressure per 4.5.2 [pd]:

$$\begin{aligned} &= \max(0.7 \cdot p_z, K_d \cdot K_a \cdot K_c \cdot 0.6 \cdot V_z^2) \\ &= \max(0.7 \cdot 1983.503, 1.0 \cdot 0.94 \cdot 1.0 \cdot 0.6 \cdot 57.5^2) \\ &= 1862.8 \text{ N/m}^2 \end{aligned}$$

Element Wind Load (operating condition):

$$\begin{aligned} &= \text{Wind Area} \cdot C_f \cdot p_d \\ &= 4.9 \cdot 0.739 \cdot 1862.824 \\ &= 6704.1 \text{ N} \end{aligned}$$

From	Height mm.	k1	k2	k3	Vz m/sec	Cf
10	1300.00	1.0000	1.0000	1.0000	57.50	0.7387
20	2325.00	1.0000	1.0000	1.0000	57.50	0.7387
30	3750.00	1.0000	1.0000	1.0000	57.50	0.7387
40	5850.00	1.0000	1.0000	1.0000	57.50	0.7387
50	6842.24	1.0000	1.0000	1.0000	57.50	0.7387

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
D - Average outer insulated diameter of vessel, mm.
Df - Damping Factor < 0.75 Unstable, > 0.95 Stable



Dr - Average outer insulated of top half of vessel, mm.
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm.
 Lc - Total length of conical section(s) of vessel, mm.
 tb - Uncorroded plate thickness at bottom of vessel, mm.
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kgf
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kgf
 Z - Maximum amplitude of vibration at top of vessel, mm.
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, $\leq 0.320E-06$ (High); $0.320E-06 < 0.400E-06$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: $0.0/6300.0 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[7.2/20.67^2] \cdot 10^4 = 168.472$ [Geometry Violation]

Compute the vibration possibility. If $V_p > 0.400E-06$ no chance. [Vp]:

= $W / (L \cdot Dr^2)$

= $17456 / (6300.0 \cdot 2193.301^2)$

= $0.57597E-06$

Since V_p is $> 0.400E-06$ no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf	Cf
PLAT 1	43200.00	2350.00	190.04	820.96	1.20
PLAT 1	93225.60	7450.00	190.04	1771.62	1.20

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf
skirt Opening	0.00	1300.00	190.04	0.00
vortex breake	0.00	1850.00	190.04	0.00
DeflectorN1B	0.00	4450.00	190.04	0.00
DeflectorN1A	0.00	4450.00	190.04	0.00
Manway Davit	0.00	3100.00	190.04	0.00
Ladder rung	0.00	4450.00	190.04	0.00
demister	0.00	6250.00	190.04	0.00
clit Wt	0.00	6250.00	190.04	0.00
Lifting Wt	0.00	6350.00	190.04	0.00

The Natural Frequency for the Vessel (Ope...) is 12.414 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm^2	Wind Pressure Kgs/m^2	Element Wind Load Kgf
10	20	1300	2436	48720	190.036	752.044
20	30	2325	2721.6	1360.8	190.036	21.0054
30	40	3750	2724	76272	190.036	1743.25
40	50	5850	2724	38136	190.036	588.669
50	60	6842.24	2721.6	14912.8	190.036	2352.08



The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Input Values:

Seismic Design Code	Spectrum
Range Type	Period
Ordinate Type	Acceleration
Include Missing Mass Components	Yes
Shock Scale Factor X Direction	1.0000
Shock Scale Factor Y Direction	0.6700
Combination Method	Group
Spectrum Name	User Defined

User Defined Spectrum Data Points:

Acceleration	Period
0.0700	0.0100
0.1170	0.0500
0.1750	0.1000
0.1750	0.1300
0.1750	0.2000
0.1750	0.2700
0.1750	0.3000
0.1750	0.4000
0.1400	0.5000
0.1270	0.5500
0.1170	0.6000
0.1050	0.6700
0.0870	0.8000
0.0700	1.0000
0.0580	1.2000
0.0500	1.4000
0.0440	1.6000
0.0390	1.8000
0.0350	2.0000
0.0290	2.4000

Computed Natural Frequencies (OPE): 4

Mode	Freq (Hz)	Freq (Rad/Sec)	Period (Sec)
1	12.4140	77.9995	0.0806
2	66.4189	417.3223	0.0151
3	68.6278	431.2014	0.0146
4	153.3376	963.4484	0.0065

Mass Participation Factors: 4

Mode	X	Y
1	0.08681693	0.00000000
2	0.00070441	0.00000000
3	-0.00000000	0.00102679
4	0.00001202	0.00000000

Computed EigenVectors, Mass Normalized: 4

Node	Dx	Dy	Rz
Mode # 1 Freq (HZ) = 12.4140			
10	0.000000	0.000000	-0.000000
20	0.032795	0.000000	-0.000490
30	0.033872	0.000000	-0.000493
40	0.100914	-0.000000	-0.000605
50	0.136218	-0.000000	-0.000616
60	0.137452	-0.000000	-0.000616
Mode # 2 Freq (HZ) = 66.4189			
10	0.000001	0.000000	-0.000000
20	0.160384	0.000000	0.000503
30	0.160520	0.000000	0.000522



40	0.036691	0.000000	0.001314
50	-0.060812	-0.000000	0.001452
60	-0.063964	-0.000000	0.001452

Mode # 3 Freq (HZ) = 68.6278

10	-0.000000	0.000002	0.000000
20	-0.000000	0.077362	0.000000
30	-0.000000	0.078048	0.000000
40	-0.000000	0.106418	0.000000
50	-0.000000	0.114033	0.000000
60	-0.000000	0.114140	0.000000

Mode # 4 Freq (HZ) = 153.3376

10	0.000000	0.000000	-0.000000
20	0.066808	0.000000	0.000709
30	0.065421	-0.000000	0.000723
40	-0.141399	0.000000	-0.000818
50	0.079331	-0.000000	-0.001789
60	0.084921	0.000000	-0.001789

Mass Percentages:

Included(X)	Included(Y)	Added(X)	Added(Y)
96.98	94.68	3.02	5.32

NOTE: In the following reports, "Max Contrib" displays the contribution of the Mode/Load combination having the maximum impact on the total (and names that Mode/Load combination).

Restraint Loads:

Node	Fx(Kgf)	Fy(Kgf)	Mz(Kg-m.)
10	2161.2	848.5	11288.0
Max	2148.4	847.3	11286.9
Contrib	1 (X)	3 (Y)	1 (X)

Element Forces and Moments:

Node	Fx(Kgf)	Fy(Kgf)	Mz(Kg-m.)
10	2160.9	847.5	11288.0
Max	2148.4	847.3	11286.9
Contrib	1 (X)	3 (Y)	1 (X)
20	2160.9	847.5	6996.9
Max	2148.4	847.3	6990.1
Contrib	1 (X)	3 (Y)	1 (X)
20	2108.9	807.2	6996.9
Max	2101.2	807.1	6990.1
Contrib	1 (X)	3 (Y)	1 (X)
30	2108.9	807.2	6892.4
Max	2101.2	807.1	6885.1
Contrib	1 (X)	3 (Y)	1 (X)
30	1890.6	632.6	6892.4
Max	1889.8	631.1	6885.1
Contrib	1 (X)	3 (Y)	1 (X)
40	1890.6	632.6	1602.3
Max	1889.8	631.1	1593.4
Contrib	1 (X)	3 (Y)	1 (X)
40	1129.3	340.4	1602.3
Max	1123.0	338.8	1593.4
Contrib	1 (X)	3 (Y)	1 (X)
50	1129.3	340.4	21.3
Max	1123.0	338.8	21.1
Contrib	1 (X)	3 (Y)	1 (X)
50	425.3	127.5	21.3



Max	422.8	126.9	21.1
Contrib	1 (X)	3 (Y)	1 (X)
60	425.3	127.5	0.0
Max	422.8	126.9	0.0
Contrib	1 (X)	3 (Y)	

Displacements:

Node	Dx(mm.)	Dy(mm.)	Rz(Deg)
10	0.0000	0.0000	0.0000
Max	0.0000	0.0000	0.0000
Contrib	1 (X)	3 (Y)	1 (X)
20	0.0724	0.0020	0.0024
Max	0.0723	0.0020	0.0024
Contrib	1 (X)	3 (Y)	1 (X)
30	0.0747	0.0020	0.0025
Max	0.0747	0.0020	0.0025
Contrib	1 (X)	3 (Y)	1 (X)
40	0.2225	0.0028	0.0030
Max	0.2225	0.0028	0.0030
Contrib	1 (X)	3 (Y)	1 (X)
50	0.3004	0.0030	0.0031
Max	0.3004	0.0030	0.0031
Contrib	1 (X)	3 (Y)	1 (X)
60	0.3031	0.0030	0.0031
Max	0.3031	0.0030	0.0031
Contrib	1 (X)	3 (Y)	1 (X)



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	5457.05	2160.91	25464.3	11288
20	30	2025	4705.01	2108.86	15302	6996.94
30	40	3450	4684	1890.62	15067.3	6892.39
40	50	5550	2940.75	1129.31	4392.45	1602.33
50	60	6275	2352.08	425.339	687.393	21.2674

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	5457.05	0.024716	0.091663	0.84589E-04
20	30	4705.01	0.093781	0.095909	0.85248E-04
30	40	4684	0.22569	0.37028	0.00010665
40	50	2940.75	0.44568	0.52206	0.00010948
50	60	2352.08	0.5248	0.52754	0.00010948

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	542.863	3392.89
20	30	606.509	3790.68
30	40	607.044	3794.02
40	50	607.044	3794.02
50	60	606.509	3790.68

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
Allowable deflection : 31.500 Actual deflection : 0.528 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	508.182	413.204	25587830	20836320
20	30	1205.14	1110.83	60267972	55633040
30	40	1269.21	1174.89	63472756	58842364
40	50	1269.21	1174.89	63472756	58842364
50	60	1205.14	1110.83	60267972	55633040



Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	91.0784	118.402	-93.7969	-101.809
20	30	165.596	215.276	-126.384	-128.474
30	40	165.596	215.276	-127.782	-129.705
40	50	165.596	215.276	-127.782	-129.705
50	60	165.596	215.276	-126.384	-128.474

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have been computed in the new and cold condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	65.3567	-2.97558	81.886
30	40	61.799	-2.81886	77.7604
40	50	61.799	-2.81886	77.7604
50	60	65.3567	-2.97558	81.886

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-4.28897	-4.28897	...	0.66164	0.66164
20	30	-1.46894	-1.46894	...	0.2478	0.2478
30	40	-1.30271	-1.30271	...	0.23429	0.23429
40	50	-0.62725	-0.62725	...	0.010299	0.010299
50	60	-0.39705	-0.39705	...	0.010894	0.010894

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	11.9848	5.31271	...	...
20	30	...	2.69735	1.23338	...	...
30	40	...	2.51111	1.14868	...	...
40	50	...	0.73204	0.26704	...	...
50	60	...	0.12117	0.0037489	...	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	4.42379
20	30	...	...	...	1.02354
30	40	...	...	...	0.95319
40	50	...	...	...	0.22573
50	60	...	...	...	0.0036126

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	-0.20115
20	30	...	-0.071259
30	40	...	-0.0528

40	50	...	-0.02841
50	60	...	-0.011257

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm^2	Earthquake For/Mom Corroded N./mm^2	Wind For/Mom No Corrosion N./mm^2	Earthquake For/Mom No Corrosion N./mm^2
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm^2
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...

**Stress Combination Load Cases for Vertical Vessels:****Load Case Definition Key**

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Un-Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.36	91.08	-16.94	93.80	0.0918	0.1806
20	1.48	165.60	-4.41	126.38	0.0089	0.0349
30	1.44	165.60	-4.05	127.78	0.0087	0.0317
40	0.12	165.60	-1.37	127.78	0.0007	0.0107
50		165.60	-0.53	126.38		0.0042

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.60	91.08	-9.58	93.80	0.0065	0.1021
20		165.60	-2.81	126.38		0.0222
30		165.60	-2.54	127.78		0.0199
40		165.60	-0.89	127.78		0.0070
50		165.60	-0.42	126.38		0.0033

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.36	91.08	-16.94	93.80	0.0918	0.1806
20	1.48	165.60	-4.41	126.38	0.0089	0.0349
30	1.44	165.60	-4.05	127.78	0.0087	0.0317
40	0.12	165.60	-1.37	127.78	0.0007	0.0107



50 165.60 -0.53 126.38 0.0042

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.48	91.08	-10.46	93.80	0.0163	0.1116
20		165.60	-3.02	126.38		0.0239
30	0.03	165.60	-2.74	127.78	0.0002	0.0214
40		165.60	-0.93	127.78		0.0073
50		165.60	-0.42	126.38		0.0033

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	118.40	0.00	101.81	0.0000	-0.0000
20	0.00	215.28	0.00	128.47	0.0000	-0.0000
30	0.00	215.28	0.00	129.70	0.0000	-0.0000
40	0.00	215.28	0.00	129.70	0.0000	-0.0000
50	0.00	215.28	0.00	128.47	0.0000	-0.0000

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	118.40	0.00	101.81	0.0000	-0.0000
20	0.00	215.28	0.00	128.47	0.0000	-0.0000
30	0.00	215.28	0.00	129.70	0.0000	-0.0000
40	0.00	215.28	0.00	129.70	0.0000	-0.0000
50	0.00	215.28	0.00	128.47	0.0000	-0.0000

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.36	91.08	-16.94	93.80	0.0918	0.1806
20	66.83	165.60		126.38	0.4036	
30	63.24	165.60		127.78	0.3819	
40	61.91	165.60		127.78	0.3739	
50	65.09	165.60		126.38	0.3931	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.48	91.08	-10.46	93.80	0.0163	0.1116
20	65.30	165.60		126.38	0.3943	
30	61.83	165.60		127.78	0.3734	
40	61.42	165.60		127.78	0.3709	
50	64.96	165.60		126.38	0.3923	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.36	91.08	-16.94	93.80	0.0918	0.1806
20		165.60	-7.39	126.38		0.0585
30		165.60	-6.87	127.78		0.0537
40		165.60	-4.19	127.78		0.0328
50		165.60	-3.50	126.38		0.0277

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.48	91.08	-10.46	93.80	0.0163	0.1116
20		165.60	-6.00	126.38		0.0475
30		165.60	-5.56	127.78		0.0435
40		165.60	-3.75	127.78		0.0294



50 165.60 -3.40 126.38 0.0269

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	118.40	0.00	101.81	0.0000	-0.0000
20	81.89	215.28		128.47	0.3804	
30	77.76	215.28		129.70	0.3612	
40	77.76	215.28		129.70	0.3612	
50	81.89	215.28		128.47	0.3804	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	118.40	0.00	101.81	0.0000	-0.0000
20	81.89	215.28		128.47	0.3804	
30	77.76	215.28		129.70	0.3612	
40	77.76	215.28		129.70	0.3612	
50	81.89	215.28		128.47	0.3804	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-4.95	93.80		0.0528
20	64.14	165.60		126.38	0.3873	
30	60.73	165.60		127.78	0.3667	
40	61.18	165.60		127.78	0.3695	
50	64.97	165.60		126.38	0.3923	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-3.49	93.80		0.0372
20	64.00	165.60		126.38	0.3865	
30	60.59	165.60		127.78	0.3659	
40	61.22	165.60		127.78	0.3697	
50	64.99	165.60		126.38	0.3925	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-4.95	93.80		0.0528
20	64.14	165.60		126.38	0.3873	
30	60.73	165.60		127.78	0.3667	
40	61.18	165.60		127.78	0.3695	
50	64.97	165.60		126.38	0.3923	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-4.95	93.80		0.0528
20	64.14	165.60		126.38	0.3873	
30	60.73	165.60		127.78	0.3667	
40	61.18	165.60		127.78	0.3695	
50	64.97	165.60		126.38	0.3923	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-4.95	93.80		0.0528
20		165.60	-1.72	126.38		0.0136
30		165.60	-1.54	127.78		0.0120
40		165.60	-0.64	127.78		0.0050



50 165.60 -0.41 126.38 0.0032

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		75.90	-5.15	78.16		0.0659
20	64.06	138.00		105.32	0.4642	
30	60.68	138.00		106.48	0.4397	
40	61.15	138.00		106.48	0.4432	
50	64.96	138.00		105.32	0.4707	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		75.90	-5.15	78.16		0.0659
20		138.00	-4.76	105.32		0.0452
30		138.00	-4.41	106.48		0.0414
40		138.00	-3.48	106.48		0.0327
50		138.00	-3.39	105.32		0.0322

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-4.95	93.80		0.0528
20	64.14	165.60		126.38	0.3873	
30	60.73	165.60		127.78	0.3667	
40	61.18	165.60		127.78	0.3695	
50	64.97	165.60		126.38	0.3923	

Absolute Maximum of the all of the Stress Ratio's 0.4707

Governing Element: Top D end
 Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
SKIRT	...	-4.5	0.7	...	...	+ Tens
SKIRT	...	-4.5	-0.7	...	-5.2	- Comp
Bottom D end	65.4	-1.5	0.2	...	64.1	+ Tens
Bottom D end	65.4	-1.5	-0.2	...	...	- Comp
Shell 1	61.8	-1.4	0.2	...	60.7	+ Tens
Shell 1	61.8	-1.4	-0.2	...	...	- Comp
shell 2	61.8	-0.7	...	...	61.2	+ Tens
shell 2	61.8	-0.7	...	...	...	- Comp
Top D end	65.4	-0.4	...	...	65.0	+ Tens
Top D end	65.4	-0.4	...	...	...	- Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: SKIRT

Total vessel (empty) weight	Fg	16919.379 Kgf
Head Weight below section 2-2	DeltaFg	1031.878 Kgf
Weight of vessel contents (liquid)	Ff	0.000 Kgf
Vessel Gross Weight	Fg + Ff	16919.379 Kgf
Skirt Mean Diameter	Dz	2023.500 mm.
Skirt Analysis Thickness (Corroded)	ez	6.500 mm.
Skirt Height	H	2000.000 mm.
Additional Axial Force at section 1-1	F1	0.000 Kgf
Moment at section 1-1	M1	15302.039 Kg-m.
Applied Moment - Bottom of Skirt	Mb	25464.297 Kg-m.
Weight at Bottom of Skirt	W	18071.277 Kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	2020.500 mm.
Head Corroded Thickness	eb	17.500 mm.

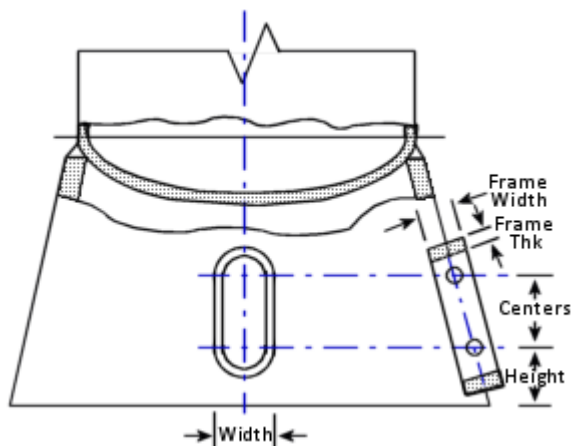
Per EN 13445 Section 7.5.4 [R, r]:

$$h_i = 501.5 \text{ mm.}$$

$$K = D_i / (2 * h_i) = 2003.0 / (2 * 501.5) = 1.997$$

$$R = D_i (0.44 * K + 0.02) = 2003.0 * (0.44 * 1.997 + 0.02) = 1800.064 \text{ mm.}$$

$$r = D_i ((0.5 / K) - 0.08) = 2003.0 * ((0.5 / 1.997) - 0.08) = 341.26 \text{ mm.}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	2114863.750 cm ⁴
Moment of Inertia	Iyy =	2114863.750 cm ⁴
Section Modulus	W =	20903048.000 mm. ³
Section Metal Area	A =	413.206 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{\min(I_{xx}, I_{yy}) / \text{Area}}$$

$$= \sqrt{\min(2114864, 2114864) / 413.206}$$

$$= 715.415 \text{ mm.}$$

Skirt Slenderness Ratio [rg/L]:



$$\begin{aligned}
 &= r_g / L \\
 &= 715.415/2000.0 \\
 &= 0.358
 \end{aligned}$$

Correct the Moment [M4]:

top of skirt is the critical end

Controlling Moment [M4] = 15302.039 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$\begin{aligned}
 &= -M / Z - W / A \\
 &= -15302.039/83612192 - 18071.277/413.20581 \\
 &= -6.084 \text{ N./mm}^2
 \end{aligned}$$

Upwind Side:

$$\begin{aligned}
 &= +(M) / Z - W / A \\
 &= +15302.039/83612192 - 18071.277/413.20581 \\
 &= -2.494 \text{ N./mm}^2
 \end{aligned}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$\begin{aligned}
 &= F_1 - F_g - F_f + 4 \cdot M_1 / D_z \\
 &= 0.0 - 16919.379 - 0.0 + 4 \cdot 15302.039/2023.5 \\
 &= 13328.7 \text{ Kgf}
 \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned}
 &= F_1 - F_g - F_f - 4 \cdot M_1 / D_z \\
 &= 0.0 - 16919.379 - 0.0 - 4 \cdot 15302.039/2023.5 \\
 &= -47167.4 \text{ Kgf}
 \end{aligned}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$\begin{aligned}
 &= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\
 &= (13328.682 + 1031.878 + 0.0) / (\pi \cdot 2020.5 \cdot 17.5) + 23.0 \cdot 2020.5 / (4 \cdot 17.5) \\
 &= 67.66 \text{ N./mm}^2
 \end{aligned}$$

Stress [σ_{1qm}]:

$$\begin{aligned}
 &= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\
 &= (-47167.438 + 1031.878 + 0.0) / (\pi \cdot 2020.5 \cdot 17.5) + 23.0 \cdot 2020.5 / (4 \cdot 17.5) \\
 &= 62.32 \text{ N./mm}^2
 \end{aligned}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$\begin{aligned}
 &= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\
 &= (0.0 + 1031.878) / (\pi \cdot 2020.5 \cdot 17.5) + 23.0 \cdot 2020.5 / (4 \cdot 17.5) \\
 &= 66.48 \text{ N./mm}^2
 \end{aligned}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$\begin{aligned}
 &= F_{zp} / (\pi \cdot D_z \cdot e_z) \\
 &= 13328.682 / (\pi \cdot 2023.5 \cdot 6.5) \\
 &= 3.16 \text{ N./mm}^2
 \end{aligned}$$

Stress [σ_{3qm}]:

$$\begin{aligned}
 &= F_{zq} / (\pi \cdot D_z \cdot e_z) \\
 &= -47167.438 / (\pi \cdot 2023.5 \cdot 6.5) \\
 &= -11.19 \text{ N./mm}^2
 \end{aligned}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:

$$\begin{aligned}
 &= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b) \\
 &= 1 - 0.5(2020.5 + 17.5 - 2023.5 + 6.5) / (341.26 + 17.5) \\
 &= 0.9707
 \end{aligned}$$



$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.971) = 13.896 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(17.5^2 + 6.5^2 + 2 \cdot 17.5 \cdot 6.5 \cdot 0.971)}$$

$$= 11.93 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 11.93 \cdot 13328.7 = 159.02 \text{ Kg-m.}$$

$$M_q = a \cdot F_{zq} = 11.93 \cdot -47167.4 = -562.74 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2$$

$$= 0.63 - 0.057 \left(\frac{17.5}{6.5} \right)^2$$

$$= 0.217$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.217 \cdot 6 \cdot 159.02 / (\pi \cdot 2020.5 \cdot 17.5^2)$$

$$= 1.04 \text{ N./mm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.217 \cdot 6 \cdot -562.739 / (\pi \cdot 2020.5 \cdot 17.5^2)$$

$$= -3.69 \text{ N./mm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.217 \cdot 6 \cdot 159.02 / (\pi \cdot 2023.5 \cdot 6.5^2)$$

$$= 7.55 \text{ N./mm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.217 \cdot 6 \cdot -562.739 / (\pi \cdot 2023.5 \cdot 6.5^2)$$

$$= -26.73 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 0.689/79.547 = 0.0087$$

$$y = 125 \cdot eb/D_b = 125 \cdot 17.5/2020.5 = 1.0827$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 1.083 = 3.9835$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b \left(\left(\frac{y}{y_A} \right) \cdot \alpha - 1 \right) / (4 \cdot eb)$$

$$= (23.0 + 0.0) \cdot 2020.5 \left(\left(\frac{13.9}{40.0} \right) \cdot 3.983 - 1 \right) / (4 \cdot 17.5)$$

$$= 25.49 \text{ N./mm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 67.66 - 1.044 + 25.486$$

$$= 92.10 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 67.66 + 1.044 - 25.486$$

$$= 43.22 \text{ N./mm}^2$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 62.319 - -3.693 + 25.486$$



$$= 91.50 \text{ N./mm}^2$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 62.319 + -3.693 - 25.486$$

$$= 33.14 \text{ N./mm}^2$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 66.483 + 1.044 + 25.486$$

$$= 93.01 \text{ N./mm}^2$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 66.483 - 1.044 - 25.486$$

$$= 39.95 \text{ N./mm}^2$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 66.483 + -3.693 + 25.486$$

$$= 88.28 \text{ N./mm}^2$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 66.483 - -3.693 - 25.486$$

$$= 44.69 \text{ N./mm}^2$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= 3.163 - 7.554$$

$$= -4.39 \text{ N./mm}^2$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= 3.163 + 7.554$$

$$= 10.72 \text{ N./mm}^2$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -11.194 - -26.732$$

$$= 15.54 \text{ N./mm}^2$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -11.194 + -26.732$$

$$= -37.93 \text{ N./mm}^2$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 137.997(3 - (1/1.5)) \cdot (67.66/137.997)^2$$

$$= 391.88 \text{ N./mm}^2$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 137.997(3 - (1/1.5)) \cdot (62.319/137.997)^2$$

$$= 395.23 \text{ N./mm}^2$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 137.997(3 - (1/1.5)) \cdot (66.483/137.997)^2$$

$$= 392.64 \text{ N./mm}^2$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$$

$$= 137.997(3 - (1/1.5)) \cdot (66.483/137.997)^2$$

$$= 392.64 \text{ N./mm}^2$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2) \\ &= 137.997(3 - (1/1.5) \cdot (3.163/137.997)^2) \\ &= 413.94 \text{ N./mm}^2 \end{aligned}$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2) \\ &= 137.997(3 - (1/1.5) \cdot (-11.194/137.997)^2) \\ &= 413.39 \text{ N./mm}^2 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm2)

Stress	Actual	Allowed	Result	Equation
Stot1pi	92.10	391.88	Pass	46
Stot1po	43.22	391.88	Pass	47
Stot1qi	91.50	395.23	Pass	48
Stot1qo	33.14	395.23	Pass	49
Stot2pi	93.01	392.64	Pass	50
Stot2po	39.95	392.64	Pass	51
Stot2qi	88.28	392.64	Pass	52
Stot2qo	44.69	392.64	Pass	53
Stot3pi	-4.39	413.94	Pass	54
Stot3po	10.72	413.94	Pass	55
Stot3qi	15.54	413.39	Pass	56
Stot3qo	-37.93	413.39	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	4519.420 mm.
Center of Gravity of the Insulation	4150.000 mm.
Center of Gravity of the Nozzles	3656.188 mm.
Center of Gravity of the Added Weights (Operating)	4917.359 mm.
Center of Gravity of the Added Weights (Empty)	4809.306 mm.
Center of Gravity of Bare Shell New and Cold	3636.085 mm.
Center of Gravity of Bare Shell Corroded	3661.999 mm.
Vessel CG in the Operating Condition	4147.964 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	4100.143 mm.
Vessel CG in the Test Condition	4123.377 mm.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	2029.9999	mm.
Skirt Thickness	STHK	8.0000	mm.
Skirt Internal Corrosion Allowance	SCA	1.5000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	24.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-285 C	
Basing Ope. All. Stress	BASOPE, f4	108.00	N./mm2
Basing Yield Stress		207.00	N./mm2
Inside Diameter of Basing	DI, D4	1942.0000	mm.
Outside Diameter of Basing	DOU, D5	2246.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	30.0000	mm.
Bolt Corrosion Allowance	BCA	1.5000	mm.
Root Area of a Single Bolt	Area	5.4000	cm2
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	137.30	N./mm2
Number of Bolts	RN, nb	16	
Diameter of Bolt Circle	DC, DBC	2135.0000	mm.
Bolt Allowable Shear Stress		107.873	N./mm2
Thickness of Gusset Plates	TGA, e7	10.0000	mm.
Width of Gussets at Top Plate	TWDT	102.0000	mm.
Width of Gussets at Base Plate	BWDT	102.0000	mm.
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	207.0	N./mm2
Height of Chair	h1	230.0000	mm.
Distance between Gussets	RG, b6	90.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	45.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	25.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	102.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	57.0000	mm.
External Corrosion Allowance	CA	1.5000	mm.
Dead Weight of Vessel	DW	17871.0	Kgf
Operating Weight of Vessel	ROW	18071.3	Kgf
Earthquake Moment on Basing	EQMOM	11792.0	Kg-m.
Wind Moment on Basing	WIMOM	25968.3	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Basing Analysis : Analyze Option

Basing Thickness Calculation method used : Simplified (Steel on Steel)

Calculation of Bolt Loads for Multiple Loading Conditions:

Bolt Load, Earthquake Operating Case [FB,e]:

$$= ((4 \cdot M/DC) - W)/nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 11792/2135.0) - 18071)/16$$

$$= 251.3 \text{ Kgf}$$

Bolt Load, Windload Empty Case [FB,w]:

$$= ((4 \cdot M/DC) - W)/nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 25968/2135.0) - 17871)/16$$

$$= 1923.8 \text{ Kgf}$$



Bolt Load, Windload Vortex Shedding Case [FB,v]:
 $= ((4 \cdot M/DC) - W)/nb$ per Jawad & Farr, Eq. 12.3
 $= ((4 \cdot 0/2135.0) - 18071)/16$
 $= -1129.5 \text{ Kgf}$

Calculation of Maximum Load per Bolt [W/Bolt], Wind + Dead Weight Condition:
 where: W = DW M = WIMOM

$= ((4 \cdot M/DC) - W)/nb$ per Jawad & Farr, Eq. 12.3
 $= ((4 \cdot 25968/2135.0) - 17871)/16$
 $= 1923.8 \text{ Kgf}$

Required Area for Each Bolt, Based on Max Load	1.3741	cm ²
Area Available in a Single Bolt (Corr)	5.4000	cm ²
Area Available in all the Bolts (Corr)	86.4000	cm ²
Bolt Stress Based on Simplified Analysis	34.9	N./mm ²
Allowable Bolt Stress	137.3 [Fact]	137.30 N./mm ²

Shear Stress in a Single Bolt [tb]:
 $= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 5457.0 / (5.4 \cdot 16)$
 $= 6.2 \text{ N./mm}^2$. Must be less than 107.9 N./mm².

Concrete Contact Area of Base Ring	CCA	9999.31	cm ²
Concrete Contact Section Modulus of Base Ring		0.4906E+09	mm. ³

Concrete Load (Simplified method), Wind in Operating Condition [Sc]:
 $= ((ppl/100 \cdot (Abt \cdot Sa) + W)/Cca) + M/CZ$ per Jawad & Farr Eq. 12.1
 $= (1.0(86.4 \cdot 137 + 18071)/9999.31) + 25968/4.9061E+09$
 $= 1.88 \text{ N./mm}^2$

Concrete Allowable Stress	8.75	N./mm ²
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Determine Maximum Bending Width of Basing Section [Rw1]:
 $= (Dou - SkirtOD)/2$
 $= (2246.0 - 2030.0)/2$
 $= 108.0 \text{ mm.}$

Determine Maximum Bending Width of Basing Section [Rw2]:
 $= (SkirtID - Di + 2 \cdot Sca)/2$
 $= (2014.0 - 1942.0 + 2 \cdot 1.5)/2$
 $= 37.5 \text{ mm.}$

Calculation of required Basing Thickness, (Simplified) [Tb]:
 $= \max(Rw1, Rw2) \cdot \sqrt{(3 \cdot Sc / S) + CA}$ per Jawad & Farr Eq. 12.12
 $= \max(108.0, 37.4999) \cdot \sqrt{(3 \cdot 1.883/137.997) + 1.5}$
 $= 23.3492 \text{ mm.}$

Basing Stress at given Thickness [Sb]:
 $= 3 \cdot Sc (\max[Rw1, Rw2] / (Tb - Ca))^2$
 $= 3 \cdot 1.883 (\max[108.0, 37.5] / (24.0 - 1.5))^2$
 $= 130.13$, must be less than 137.997 N./mm²

Required Thickness of Top Plate in Tension:

(Calculated as a fixed beam per Megyesy)

$Ft = (Sa \cdot Abss)$,	Bolt Allowable Stress * Area
$Rm = (Ft \cdot 2 \cdot Cg)/8$,	Bending Moment
Sb	Allowable Bending Stress
$Wt = (Topwth - Bnd)$,	Width of Section

Required Thickness of Top Plate [T]:

$= \sqrt{(6 \cdot Rm / (Sb \cdot Wt)) + CA}$
 $= \sqrt{(6 \cdot 85 / (138 \cdot 72.0)) + 1.5}$
 $= 23.943 \text{ mm.}$

Required Thickness of Continuous Top Ring per Moss:

$a = (Dc - SkirtOD)/2$	Skirt Distance to Bolt Circle
$P = Sa \cdot Abss$	Bolt Allowable Stress * Area
$l = Avgwdt$	Average Gusset Width



g1 = Gamma 1 Constant Term f(b/l)
 g2 = Gamma 2 Constant Term f(b/l)
 g = Flat distance / 2 Nut 1/2 Dimension (from Tema)
 Fb = Allowable Bending Stress

$M_o = P/(4\pi)[1.3(\ln((2l\sin(\pi a/l)/(\pi g))) + 1] - [(0.7 - g^2)P/(4\pi)]$ Moment Term

Required Thickness of Top Ring [T]:

$= \sqrt{(6 \cdot \text{Abs}(M_o) / F_b)} + CA$ Required Thickness
 $= \sqrt{(6 \cdot 25/138)} + 1.5$
 $= 21.994 \text{ mm.}$

Required Thickness of Gusset in Compression, per AISC E2-1:

1. Allowed Compression at Given Thickness:

Factor K1/r Per E2-1	62.6992	
Factor Cc Per E2-1	138.7031	
Allowable Buckling Str. per E2-1	101.85	N./mm ²
Actual Buckling Str. at Given Thickness	36.34	N./mm ²

Required Gusset/Support Plate thickness, + CA 6.7859 mm.

2. Allowed Compression at Calculated Thickness:

Factor K1/r Per E2-1	118.6165	
Factor Cc Per E2-1	138.7031	
Allowable Buckling Str. per E2-1	68.77	N./mm ²
Act. Buckling Str. at Calculated Thickness	68.76	N./mm ²

Summary of Basing Thickness Calculations:

Required Basing Thickness (simplified)	23.3492	mm.
Actual Basing Thickness	24.0000	mm.

Required Top Ring/Plate Thickness as a Fixed Beam	23.9431	mm.
Required Thickness of Continuous Top Ring (Moss)	21.9939	mm.
Actual Top Plate/Ring Thickness	25.0000	mm.

Required Gusset/Support Plate thickness, + CA	6.7859	mm.
Actual Gusset/Support Plate Thickness	10.0000	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$= \text{SkirtStress} \cdot (\text{SkirtThickness} - CA)$
 $= 16.935 \cdot (8.0 - 1.5)$
 $= 11.22 \text{ Kg/mm.}$

Results for Computed Minimum Basing Weld Size [BWeld]:

$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 11 / [(0.4 \cdot 207) \cdot 2 \cdot 0.707]$
 $= 0.94 \text{ mm.}$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$= \text{Bolt Load} / (Cmwh + 2 \cdot (Hg + Tta))$
 $= 7560.2 / (110.0 + 2 \cdot (181.0 + 25.0))$
 $= 14.483 \text{ Kg/mm.}$

Horizontal Plate Load [Wh]:

$= \text{Bolt Load} \cdot e / (Cmwh \cdot (Hg + Tta) + 0.6667 \cdot (Hg + Tta)^2)$
 $= 7560.2 \cdot 52.5 / (110.0 \cdot (206.0) + 0.6667 \cdot (206.0)^2)$
 $= 7.79 \text{ Kg/mm.}$

Resultant Weld Load [Wr]:

$= \sqrt{(Wv^2 + Wh^2)}$
 $= \sqrt{(14.48^2 + 7.79^2)}$
 $= 16.445 \text{ Kg/mm.}$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 16.45 / [(0.4 \cdot 207) \cdot 2 \cdot 0.707]$



= 1.378 mm.

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 7560.2 / (2 \cdot 102.0)$$

$$= 37.06 \text{ Kg/mm.}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$

$$= 7560.2 \cdot 52.5 / (2 \cdot 206.0 \cdot 102.0)$$

$$= 9.445 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{37.06^2 + 9.44^2}$$

$$= 38.245 \text{ Kg/mm.}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 38.24 / [(0.4 \cdot 207) \cdot 2 \cdot 0.707]$$

$$= 3.204 \text{ mm.}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	4.7625	mm.
Required Gusset to Skirt Double Fillet Weld Size	4.7625	mm.
Required Top Plate to Skirt Weld Size	6.3500	mm.
Required Gusset to Top Plate Double Fillet Weld Size	3.2035	mm.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N32"

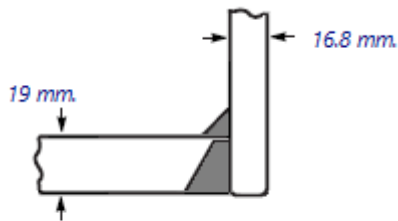
From: 20

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	19.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		19.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1250.0	mm.
Centerline Distance to Tangent		800.0	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 23.0 \cdot 2003.0 \cdot 0.8987 / (2 \cdot 138.0 \cdot 1.0 \cdot 0.2 \cdot 23.0)$
 $= 15.0273 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([23.0 / (138.0 \cdot 1.0)] - 1))$
 $= 0.4521 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3890 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	119.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	59.6640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N32".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9521 mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 18.1876 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 18.1876 mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.2340 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm.
Wall Thickness per table UG-45	tb3	= 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(18.1876, 3.0)]$
 $= 6.3000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.9521, 6.3)$
 $= 6.3000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
 (see 319.4.4 and 302.3.5):**

Sustained : 14.1, Allowable : 138.0 N./mm2 Passed



Expansion :	0.0,	Allowable :	330.9 N./mm2	Passed
Occasional :	1.6,	Allowable :	183.5 N./mm2	Passed
Shear :	2.9,	Allowable :	96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: N32"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4897 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.8647 mm.

Input Echo, WRC107/537 Item 1, Description: N32" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm.
Vessel Diameter	Dv	3600.000	mm.
Vessel Thickness	Tv	19.000	mm.
Design Temperature	T1	80.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm2
Vessel Hot S.I. Allowable	Smh	138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm.
Nozzle Diameter	Dn	50.800	mm.
Nozzle Thickness	Tn	16.764	mm.
Nozzle Material		SA-350 LF2	
Nozzle UNS Number		K03011	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm2
Design Internal Pressure	Dp	23.000	bars
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	-165.0	Kgf
Longitudinal Shear	(SUS)	(V1) V1	260.0	Kgf
Circumferential Shear	(SUS)	(Vc) V2	260.0	Kgf

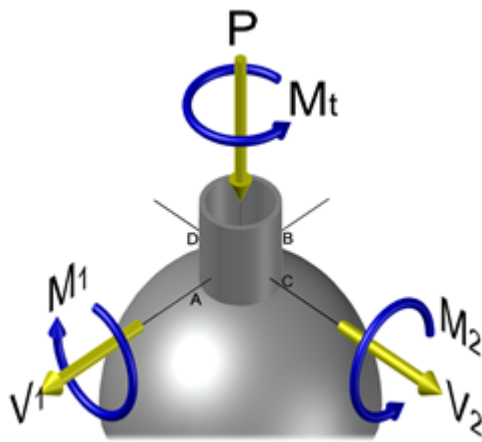


Circumferential Moment (SUS)	(Mc)	M1	35.0	Kg-m.
Longitudinal Moment (SUS)	(Ml)	M2	55.0	Kg-m.
Torsional Moment (SUS)		Mt	0.7	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(1810.25 (19.0 - 1.5))}$
 $= 671.685 \text{ mm.}$

WRC 107 Stress Calculation for SUSained loads:

Radial Load		P	-165.0	Kgf
Circumferential Shear	(VC)	V2	260.0	Kgf
Longitudinal Shear	(VL)	V1	260.0	Kgf
Circumferential Moment	(MC)	M1	35.0	Kg-m.
Longitudinal Moment	(ML)	M2	55.0	Kg-m.
Torsional Moment		MT	0.7	Kg-m.

Dimensionless Param: U = 0.24 TAU = 5.00 (2.26) RHO = 1.15

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.11888	(A,B,C,D)
$M(x) / P$	SP 2	0.15266	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / MC$	SM 2	0.29863	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / MC$	SM 2	0.57203	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / ML$	SM 2	0.29863	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / ML$	SM 2	0.57203	(A,B,C,D)
$N(y) * T / P$	SP 2	0.19065	(A,B,C,D)
$M(y) / P$	SP 2	0.06864	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / MC$	SM 2	0.02167	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / MC$	SM 2	0.31322	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / ML$	SM 2	0.02167	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / ML$	SM 2	0.31322	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Rad. Bend. P		4.8	-4.8	4.8	-4.8	4.8	-4.8	4.8	-4.8
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.9	-1.9	1.9	1.9
Rad. Bend. MC		0.0	0.0	0.0	0.0	-21.6	21.6	21.6	-21.6
Rad. Memb. ML		-3.0	-3.0	3.0	3.0	0.0	0.0	0.0	0.0
Rad. Bend. ML		-34.0	34.0	34.0	-34.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-31.4	26.8	42.4	-35.2	-18.0	15.5	29.0	-23.9
Tang. Memb. P		1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Tang. Bend. P		2.2	-2.2	2.2	-2.2	2.2	-2.2	2.2	-2.2
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC		0.0	0.0	0.0	0.0	-11.8	11.8	11.8	-11.8
Tang. Memb. ML		-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Tang. Bend. ML		-18.6	18.6	18.6	-18.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-15.6	17.2	22.0	-19.6	-8.8	10.5	15.2	-12.9
Shear VC		1.1	1.1	-1.1	-1.1	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		1.1	1.1	-1.1	-1.1	-1.1	-1.1	1.1	1.1
Str. Int.		31.5	26.9	42.4	35.3	18.1	15.7	29.1	24.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4
Rad. Pl (SUS)		-2.3	-2.3	3.6	3.6	-1.3	-1.3	2.5	2.5
Rad. Q (SUS)		-29.1	29.1	38.8	-38.8	-16.8	16.8	26.5	-26.5
Long. Pm (SUS)		118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4
Long. Pl (SUS)		0.8	0.8	1.2	1.2	0.9	0.9	1.1	1.1
Long. Q (SUS)		-16.4	16.4	20.8	-20.8	-9.7	9.7	14.0	-14.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.1	1.1	-1.1	-1.1	-1.1	-1.1	1.1	1.1
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4
Pm+Pl (SUS)		119.5	119.5	122.4	122.4	119.7	119.7	121.5	121.5
Pm+Pl+Q (Total)		102.9	145.3	160.8	98.9	109.7	134.1	147.5	105.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	118.40	138.00	Passed
Pm+Pl (SUS)	122.41	207.00	Passed
Pm+Pl+Q (TOTAL)	160.84	413.99	Passed



*Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.*

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Input, Nozzle Desc: N1A12"

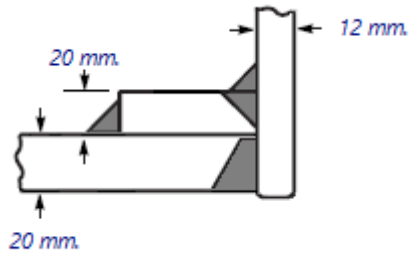
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2000.00	mm.
Design Length of Section	L	4633.3325	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2050.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	201.4760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	560.0000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	16.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		115.6000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N1A12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	12.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 16.8616 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 153.9 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 2.5911 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8534 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	615.6000 mm.
Parallel to Vessel Wall, opening length	d	307.8000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.2500 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

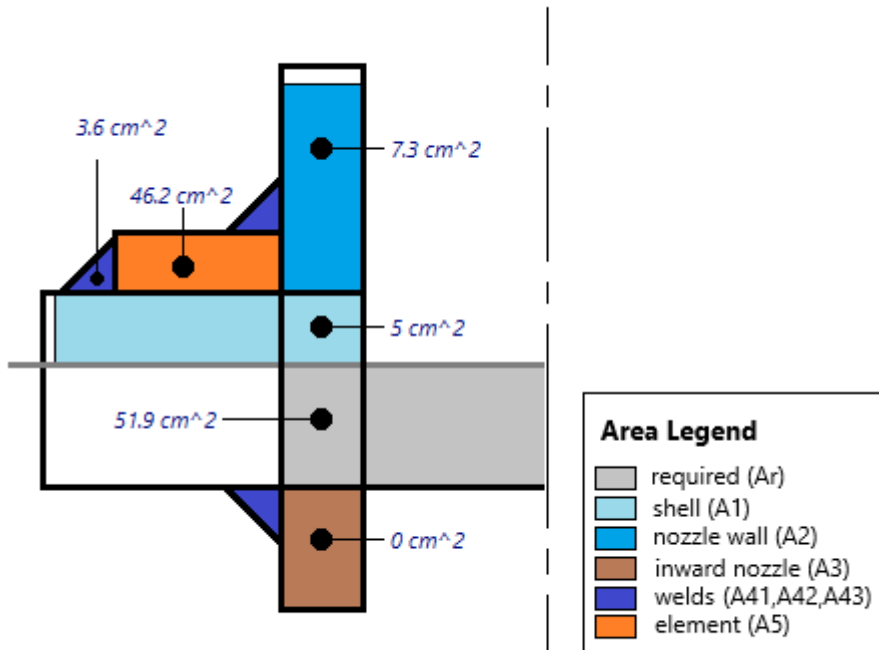


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	51.900	14.028	NA
Area in Shell	A1	5.043	28.886	NA
Area in Nozzle Wall	A2	7.316	8.923	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.560	3.560	NA
Area in Element	A5	46.240	46.240	NA
TOTAL AREA AVAILABLE	Atot	62.159	87.609	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 508.7073 20.0000 mm.
 Based on given Pad Diameter: 560.0000 15.5629 mm.
 Based on the Estimated Diameter Limit: 614.0125 12.6157 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (307.8 \cdot 16.8616 \cdot 1.0 + 2 \cdot 10.5 \cdot 16.8616 \cdot 1.0 \cdot (1 - 1.0))$
 $= 51.900 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 307.8(1.0 \cdot 18.5 - 1.0 \cdot 16.862) - 2 \cdot 10.5$
 $(1.0 \cdot 18.5 - 1.0 \cdot 16.8616) \cdot (1 - 1.0)$
 $= 5.043 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 46.25) \cdot (10.5 - 2.59) \cdot 1.0$



= 7.316 cm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 16.0^2 \cdot 1.0$
= 3.560 cm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
= $(560.0 - 328.8) \cdot 20.0 \cdot 1.0$
= 46.240 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.0911 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 18.3616 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 18.3616 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.2357 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 9.8312 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[9.831, max(18.3616, 3.0)]
= 9.8312 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(4.0911, 9.8312)
= 9.8312 mm.

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 37.8,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 307.2 N./mm ²	Passed
Occasional	: 16.3,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 14.9,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -38 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -38 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 23.00/51.00 = 0.451

Weld Size Calculations, Description: N1A12"

Intermediate Calc. for nozzle/shell Welds Tmin 10.5000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 18.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.2500 = 0.5*TminPad	11.3120 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (51.9001 - 5.0429 + 2 \cdot 10.5 \cdot 1.0 \cdot (1.0 \cdot 18.5 - 16.8616)) \cdot 138)$
 $= 66419.31 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (7.3157 + 46.24 + 3.56 - 0.0 \cdot 1.0) \cdot 138$
 $= 80370.48 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (7.3157 + 0.0 + 1. + (3.885)) \cdot 138$
 $= 17168.26 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (7.3157 + 0.0 + 3.56 + 46.24 + (3.885)) \cdot 138$
 $= 85837.27 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 328.8 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 35611. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 560.0 \cdot 16.0 \cdot 0.49 \cdot 138$
 $= 97043. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 159.15) \cdot (12.0 - 1.5) \cdot 0.7 \cdot 138$
 $= 51711. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 328.8 \cdot 20.0 \cdot 0.74 \cdot 138$
 $= 107561. \text{ Kgf}$



Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 328.8 \cdot (20.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 99494. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (97043 + 51711) = 148754 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (35611 + 107561 + 99494 + 0) = 242666 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (97043 + 99494 + 0) = 196537 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 148754 Kgf, must exceed W = 66419 Kgf or W1 = 80370 Kgf
 Path 2-2 = 242666 Kgf, must exceed W = 66419 Kgf or W2 = 17168 Kgf
 Path 3-3 = 196537 Kgf, must exceed W = 66419 Kgf or W3 = 85837 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 13.6062 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 235.0823 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 3.788 %

Input Echo, WRC297 Item 1, Description: N1A12" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2000.000	mm.
Shell Thickness	20.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	304.800	mm.
Nozzle Thickness	12.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	560.000	mm.
Thickness of Reinforcing Pad	20.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTained loads.



Design Internal Pressure	Dp	23.00	bars
Radial Load	P	-830.00	Kgf
Circumferential Shear	Vc	1225.00	Kgf
Longitudinal Shear	Vl	1225.00	Kgf
Circumferential Moment	Mc	1105.00	Kg-m.
Longitudinal Moment	Ml	1469.99	Kg-m.
Torsional Moment	Mt	1840.01	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 328.8 + 2 \cdot 1.65 \cdot \sqrt{(1020.75(20.0 - 1.5))}$
 $= 782.281 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2041.500 mm.
Nozzle Outside Diameter	(d) =	328.800 mm.
Vessel Thickness used	(T) =	38.500 mm.
Nozzle Thickness used	(t) =	10.500 mm.
T / t	=	3.667
d / t	=	31.314

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.173$$

Nr/P	=	0.072
Mr/P	=	0.078
M0/P	=	0.090
N0/P	=	0.245
MrD/Mc	=	0.142
NrDL/Mc	=	0.109
M0d/Mc	=	0.211
N0DL/Mc	=	0.194
MrD/Ml	=	0.112
NrDL/Ml	=	0.091
M0D/Ml	=	0.151
N0DL/Ml	=	0.378

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-11	-11	11	11
Outplane Bending (Ml)	-26	26	26	-26
Normal Pressure Stress	58	61	58	61
Outplane Stress Summary	25	74	100	43

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (Ml)	-2	-2	2	2
Inplane Bending (Ml)	-19	19	19	-19



Inplane Pressure Stress	29	29	29	29
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Inplane Stress Summary	9	44	54	10
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress	25	75	101	43
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-28	28	28	-28
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	29	29	29	29

Outplane Stress Summary	1	51	66	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-18	18	18	-18
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	61	58	61

Inplane Stress Summary	40	75	82	42
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	40	75	83	42
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-11	-11	11	11
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	39	41	39	41
Hoop Stress Summary	29	32	51	54

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	18	-18	18	-18
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-157	157	157	-157
Axial Pressure Stress	19	19	19	19
Axial Stress Summary	-136	141	214	-138

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	12	12	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	167	142	214	193

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	39	41	39	41
Hoop Stress Summary	36	38	44	47

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	18	-18	18	-18
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-156	156	156	-156



Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	19	19	19	19
Axial Stress Summary	-131	145	209	-141

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10
Shear Stress Summary	7	7	12	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	168	145	210	190

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 328.8 + 2 \cdot 1.65 \cdot \sqrt{(1010.75(20.0 - 1.5))}$
 $= 780.055 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	2021.500 mm.
Nozzle Outside Diameter (d)	=	560.000 mm.
Vessel Thickness used (T)	=	18.500 mm.
Nozzle Thickness used (t)	=	126.100 mm.
T / t	=	0.147
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.896$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.062
Mr/P	=	0.110
M0/P	=	0.032
N0/P	=	0.052
MrD/Mc	=	0.270
NrDL/Mc	=	0.102
M0d/Mc	=	0.081
N0DL/Mc	=	0.073
MrD/Ml	=	0.084
NrDL/Ml	=	0.077
M0D/Ml	=	0.024
N0DL/Ml	=	0.066

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	123	125	123	125
Outplane Stress Summary	113	128	145	116



Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-37	37	37	-37
Inplane Pressure Stress	61	61	61	61
Inplane Stress Summary	35	79	122	15

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	113	128	145	116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-27	27	27	-27
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	61	61	61	61
Outplane Stress Summary	35	81	99	34

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-91	91	91	-91
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	123	125	123	125
Inplane Stress Summary	43	197	237	25

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0



Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	43	197	237	35

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	58	61	58	61	58	61	58	61
Circ. Pl (SUS)	-10	-10	12	12	-2	-2	2	2
Circ. Q (SUS)	-24	24	28	-28	-16	16	20	-20
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	-2	-2	2	2	-3	-3	5	5
Long. Q (SUS)	-17	17	21	-21	-26	26	30	-30
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	58.0	61.0	58.0	61.0	58.0	61.0	58.0	61.0
Pm+Pl (SUS)	48.0	51.0	70.0	73.0	56.0	59.0	60.0	63.0
Pm+Pl+Q (Total)	24.3	75.1	98.1	45.1	40.2	75.2	80.2	43.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	61.00	138.00	Passed
Pm+Pl (SUS)	73.00	207.00	Passed
Pm+Pl+Q (TOTAL)	98.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	123	125	123	125	123	125	123	125
Circ. Pl (SUS)	-3	-3	5	5	-4	-4	6	6
Circ. Q (SUS)	-6	6	14	-14	-76	76	106	-106
Long. Pm (SUS)	61	61	61	61	61	61	61	61
Long. Pl (SUS)	-4	-4	6	6	-3	-3	5	5
Long. Q (SUS)	-22	22	52	-52	-23	23	31	-31
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	123.0	125.0	123.0	125.0	123.0	125.0	123.0	125.0

Pm+Pl (SUS)	120.0	122.0	128.0	130.0	119.0	121.0	129.0	131.0
Pm+Pl+Q (Total)	114.0	128.0	142.0	116.0	43.1	197.0	235.0	35.1

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	125.00	138.00	Passed
Pm+Pl (SUS)	131.00	207.00	Passed
Pm+Pl+Q (TOTAL)	235.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	39	41	39	41	39	41	39	41
Circ. Pl (SUS)	-10	-10	12	12	-3	-3	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	-17	-17	17	17	-13	-13	13	13
Long. Q (SUS)	-139	139	175	-175	-138	138	174	-174
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	39.0	41.0	39.0	41.0	39.0	41.0	39.0	41.0
Pm+Pl (SUS)	29.1	31.1	51.3	53.2	36.1	38.1	44.3	46.3
Pm+Pl+Q (Total)	167.7	142.3	211.4	192.7	168.8	144.6	206.9	189.5

Nozzle Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.00	138.00	Passed
Pm+Pl (SUS)	53.23	207.00	Passed
Pm+Pl+Q (TOTAL)	211.40	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N1B12"

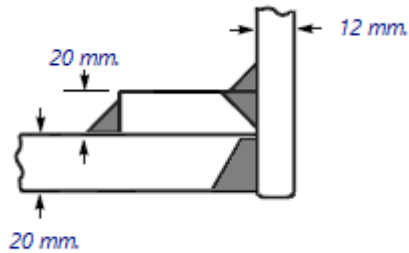
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2000.00	mm.
Design Length of Section	L	4633.3325	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2050.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	201.4760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	560.0000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	16.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		115.6000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N1B12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	12.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 16.8616 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 153.9 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 2.5911 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8534 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	615.6000 mm.
Parallel to Vessel Wall, opening length	d	307.8000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.2500 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

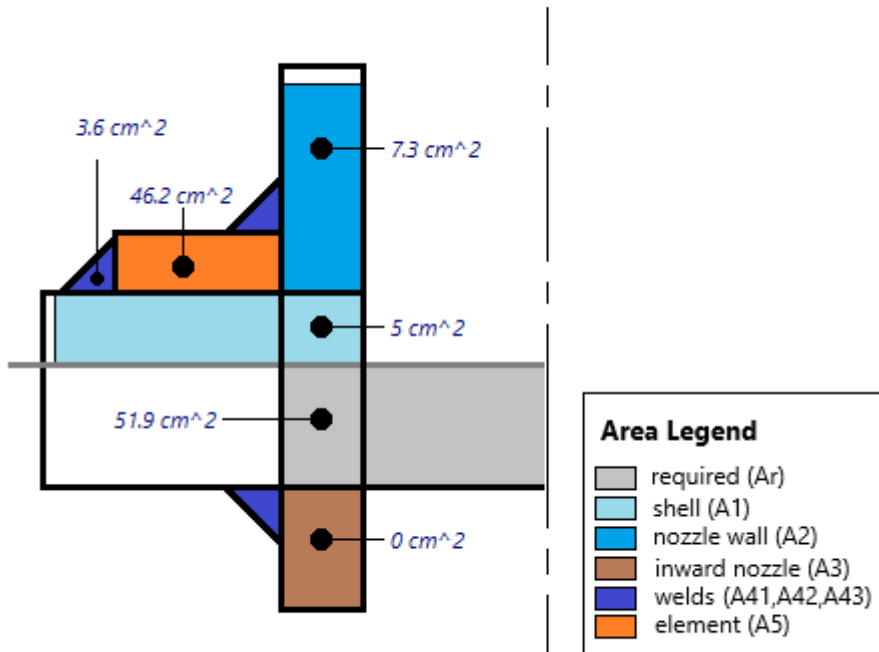


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	51.900	14.028	NA
Area in Shell	A1	5.043	28.886	NA
Area in Nozzle Wall	A2	7.316	8.923	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.560	3.560	NA
Area in Element	A5	46.240	46.240	NA
TOTAL AREA AVAILABLE	Atot	62.159	87.609	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 508.7073 20.0000 mm.
 Based on given Pad Diameter: 560.0000 15.5629 mm.
 Based on the Estimated Diameter Limit: 614.0125 12.6157 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (307.8 \cdot 16.8616 \cdot 1.0 + 2 \cdot 10.5 \cdot 16.8616 \cdot 1.0 \cdot (1 - 1.0))$
 $= 51.900 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 307.8(1.0 \cdot 18.5 - 1.0 \cdot 16.862) - 2 \cdot 10.5$
 $(1.0 \cdot 18.5 - 1.0 \cdot 16.8616) \cdot (1 - 1.0)$
 $= 5.043 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 46.25) \cdot (10.5 - 2.59) \cdot 1.0$



= 7.316 cm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 16.0^2 \cdot 1.0$
= 3.560 cm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
= $(560.0 - 328.8) \cdot 20.0 \cdot 1.0$
= 46.240 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.0911 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 18.3616 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 18.3616 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.2357 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 9.8312 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[9.831, max(18.3616, 3.0)]
= 9.8312 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(4.0911, 9.8312)
= 9.8312 mm.

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 37.8,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 307.2 N./mm ²	Passed
Occasional	: 16.3,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 14.9,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -38 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -38 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2) -85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = 23.00/51.00 = 0.451

Weld Size Calculations, Description: N1B12"

Intermediate Calc. for nozzle/shell Welds Tmin 10.5000 mm.
Intermediate Calc. for pad/shell Welds TminPad 18.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.2500 = 0.5*TminPad	11.3120 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
= max(0, (51.9001 - 5.0429 + 2 * 10.5 * 1.0 *
(1.0 * 18.5 - 16.8616))138)
= 66419.31 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)*fr2)*Sv
= (7.3157 + 46.24 + 3.56 - 0.0 * 1.0) * 138
= 80370.48 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 * tn * t * fr1)) * Sv
= (7.3157 + 0.0 + 1. + (3.885)) * 138
= 17168.26 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2*tn*t*fr1))*S
= (7.3157 + 0.0 + 3.56 + 46.24 + (3.885)) * 138
= 85837.27 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) * Dlo * Wo * 0.49 * Snw
= (3.1416/2.0) * 328.8 * 10.0 * 0.49 * 138
= 35611. Kgf

Shear, Pad Element Weld [Spew]:
= (π/2) * DP * WP * 0.49 * SEW
= (3.1416/2.0) * 560.0 * 16.0 * 0.49 * 138
= 97043. Kgf

Shear, Nozzle Wall [Snw]:
= (π * (Dlr + Dlo)/4) * (Thk - Can) * 0.7 * Sn
= (3.1416 * 159.15) * (12.0 - 1.5) * 0.7 * 138
= 51711. Kgf

Tension, Pad Groove Weld [Tpgw]:
= (π/2) * Dlo * Wgpn * 0.74 * Seg
= (3.1416/2) * 328.8 * 20.0 * 0.74 * 138
= 107561. Kgf



Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 328.8 \cdot (20.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 99494. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (97043 + 51711) = 148754 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (35611 + 107561 + 99494 + 0) = 242666 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (97043 + 99494 + 0) = 196537 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 148754 Kgf, must exceed W = 66419 Kgf or W1 = 80370 Kgf
 Path 2-2 = 242666 Kgf, must exceed W = 66419 Kgf or W2 = 17168 Kgf
 Path 3-3 = 196537 Kgf, must exceed W = 66419 Kgf or W3 = 85837 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 13.6062 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 235.0823 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.788 %

Input Echo, WRC297 Item 1, Description: N1B12" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2000.000	mm.
Shell Thickness	20.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	304.800	mm.
Nozzle Thickness	12.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	560.000	mm.
Thickness of Reinforcing Pad	20.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.



Design Internal Pressure	Dp	23.00	bars
Radial Load	P	-830.00	Kgf
Circumferential Shear	Vc	1225.00	Kgf
Longitudinal Shear	Vl	1225.00	Kgf
Circumferential Moment	Mc	1105.00	Kg-m.
Longitudinal Moment	Ml	1469.99	Kg-m.
Torsional Moment	Mt	1840.01	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 328.8 + 2 \cdot 1.65 \cdot \sqrt{(1020.75(20.0 - 1.5))}$
 $= 782.281 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2041.500 mm.
Nozzle Outside Diameter	(d) =	328.800 mm.
Vessel Thickness used	(T) =	38.500 mm.
Nozzle Thickness used	(t) =	10.500 mm.
T / t	=	3.667
d / t	=	31.314

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.173$$

Nr/P	=	0.072
Mr/P	=	0.078
M0/P	=	0.090
N0/P	=	0.245
MrD/Mc	=	0.142
NrDL/Mc	=	0.109
M0d/Mc	=	0.211
N0DL/Mc	=	0.194
MrD/Ml	=	0.112
NrDL/Ml	=	0.091
M0D/Ml	=	0.151
N0DL/Ml	=	0.378

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-11	-11	11	11
Outplane Bending (Ml)	-26	26	26	-26
Normal Pressure Stress	58	61	58	61
Outplane Stress Summary	25	74	100	43

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (Ml)	-2	-2	2	2
Inplane Bending (Ml)	-19	19	19	-19



Inplane Pressure Stress	29	29	29	29
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Inplane Stress Summary	9	44	54	10
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	25	75	101	43
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-28	28	28	-28
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	29	29	29	29

Outplane Stress Summary	1	51	66	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-18	18	18	-18
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	61	58	61

Inplane Stress Summary	40	75	82	42
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	40	75	83	42
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-11	-11	11	11
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	39	41	39	41
Hoop Stress Summary	29	32	51	54

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	18	-18	18	-18
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-157	157	157	-157
Axial Pressure Stress	19	19	19	19
Axial Stress Summary	-136	141	214	-138

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	12	12	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	167	142	214	193

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	39	41	39	41
Hoop Stress Summary	36	38	44	47

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	18	-18	18	-18
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-156	156	156	-156



Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	19	19	19	19
Axial Stress Summary	-131	145	209	-141

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10
Shear Stress Summary	7	7	12	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	168	145	210	190

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 328.8 + 2 \cdot 1.65 \cdot \sqrt{(1010.75(20.0 - 1.5))}$
 $= 780.055 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	2021.500 mm.
Nozzle Outside Diameter (d)	=	560.000 mm.
Vessel Thickness used (T)	=	18.500 mm.
Nozzle Thickness used (t)	=	126.100 mm.
T / t	=	0.147
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.896$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.062
Mr/P	=	0.110
M0/P	=	0.032
N0/P	=	0.052
MrD/Mc	=	0.270
NrDL/Mc	=	0.102
M0d/Mc	=	0.081
N0DL/Mc	=	0.073
MrD/Ml	=	0.084
NrDL/Ml	=	0.077
M0D/Ml	=	0.024
N0DL/Ml	=	0.066

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	123	125	123	125
Outplane Stress Summary	113	128	145	116



Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-37	37	37	-37
Inplane Pressure Stress	61	61	61	61
Inplane Stress Summary	35	79	122	15

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	113	128	145	116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-27	27	27	-27
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	61	61	61	61
Outplane Stress Summary	35	81	99	34

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-91	91	91	-91
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	123	125	123	125
Inplane Stress Summary	43	197	237	25

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0



Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	43	197	237	35

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	58	61	58	61	58	61	58	61
Circ. Pl (SUS)	-10	-10	12	12	-2	-2	2	2
Circ. Q (SUS)	-24	24	28	-28	-16	16	20	-20
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	-2	-2	2	2	-3	-3	5	5
Long. Q (SUS)	-17	17	21	-21	-26	26	30	-30
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	58.0	61.0	58.0	61.0	58.0	61.0	58.0	61.0
Pm+Pl (SUS)	48.0	51.0	70.0	73.0	56.0	59.0	60.0	63.0
Pm+Pl+Q (Total)	24.3	75.1	98.1	45.1	40.2	75.2	80.2	43.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	61.00	138.00	Passed
Pm+Pl (SUS)	73.00	207.00	Passed
Pm+Pl+Q (TOTAL)	98.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	123	125	123	125	123	125	123	125
Circ. Pl (SUS)	-3	-3	5	5	-4	-4	6	6
Circ. Q (SUS)	-6	6	14	-14	-76	76	106	-106
Long. Pm (SUS)	61	61	61	61	61	61	61	61
Long. Pl (SUS)	-4	-4	6	6	-3	-3	5	5
Long. Q (SUS)	-22	22	52	-52	-23	23	31	-31
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	123.0	125.0	123.0	125.0	123.0	125.0	123.0	125.0

Pm+Pl (SUS)	120.0	122.0	128.0	130.0	119.0	121.0	129.0	131.0
Pm+Pl+Q (Total)	114.0	128.0	142.0	116.0	43.1	197.0	235.0	35.1

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	125.00	138.00	Passed
Pm+Pl (SUS)	131.00	207.00	Passed
Pm+Pl+Q (TOTAL)	235.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	39	41	39	41	39	41	39	41
Circ. Pl (SUS)	-10	-10	12	12	-3	-3	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	-17	-17	17	17	-13	-13	13	13
Long. Q (SUS)	-139	139	175	-175	-138	138	174	-174
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	39.0	41.0	39.0	41.0	39.0	41.0	39.0	41.0
Pm+Pl (SUS)	29.1	31.1	51.3	53.2	36.1	38.1	44.3	46.3
Pm+Pl+Q (Total)	167.7	142.3	211.4	192.7	168.8	144.6	206.9	189.5

Nozzle Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.00	138.00	Passed
Pm+Pl (SUS)	53.23	207.00	Passed
Pm+Pl+Q (TOTAL)	211.40	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: MH124"

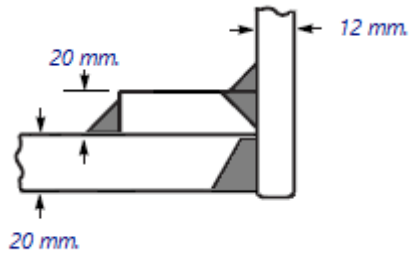
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2000.00	mm.
Design Length of Section	L	4633.3325	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		800.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	213.3760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1140.0000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	16.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		253.2000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: MH124"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 16.8616 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 306.3 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 5.1570 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 1.2829 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1225.2000 mm.
Parallel to Vessel Wall, opening length	d	612.6000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.2500 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

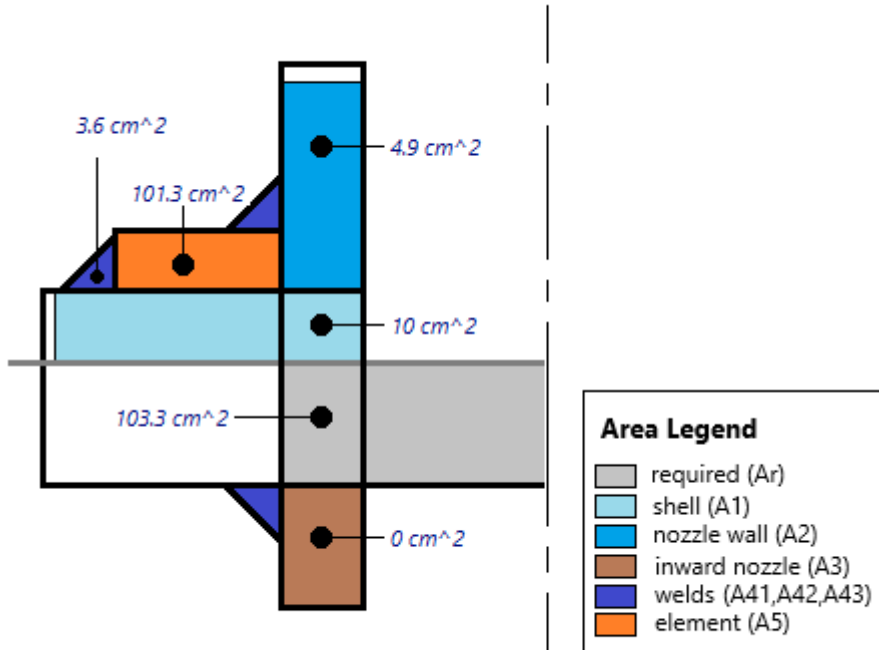


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	103.294	27.920	NA
Area in Shell	A1	10.037	57.491	NA
Area in Nozzle Wall	A2	4.942	8.526	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.560	3.560	NA
Area in Element	A5	101.280	101.280	NA
TOTAL AREA AVAILABLE	Atot	119.819	170.857	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1057.3770 20.0000 mm.
 Based on given Pad Diameter: 1140.0000 16.7368 mm.
 Based on the Estimated Diameter Limit: 1223.6125 14.3650 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (612.6 \cdot 16.8616 \cdot 1.0 + 2 \cdot 10.5 \cdot 16.8616 \cdot 1.0 \cdot (1 - 1.0))$
 $= 103.294 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 612.6(1.0 \cdot 18.5 - 1.0 \cdot 16.862) - 2 \cdot 10.5$
 $(1.0 \cdot 18.5 - 1.0 \cdot 16.8616) \cdot (1 - 1.0)$
 $= 10.037 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 46.25) \cdot (10.5 - 5.16) \cdot 1.0$



= 4.942 cm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 16.0^2 \cdot 1.0$
= 3.560 cm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(1140.0 - 633.6) \cdot 20.0 \cdot 1.0$
= 101.280 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 6.6570$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= t_a
= 6.6570 mm.

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	:	-67 °C
Gov. MDMT of the Reinforcement Pad	:	-38 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-38 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = $23.00/51.00 = 0.451$

Weld Size Calculations, Description: MH124"

Intermediate Calc. for nozzle/shell Welds	T_{min}	10.5000 mm.
Intermediate Calc. for pad/shell Welds	T_{minPad}	18.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$9.2500 = 0.5 * T_{minPad}$	$11.3120 = 0.7 * W_p$ mm.



Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (103.2943 - 10.0367 + 2 \cdot 10.5 \cdot 1.0 \cdot (1.0 \cdot 18.5 - 16.8616)) \cdot 138) \\ &= 131711.84 \text{ Kgf} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\ &= (4.9423 + 101.28 + 3.56 - 0.0 \cdot 1.0) \cdot 138 \\ &= 154480.34 \text{ Kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (4.9423 + 0.0 + 1. + (3.885)) \cdot 138 \\ &= 13828.49 \text{ Kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (4.9423 + 0.0 + 3.56 + 101.28 + (3.885)) \cdot 138 \\ &= 159947.12 \text{ Kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 633.6 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 68623. \text{ Kgf} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\ &= (3.1416/2.0) \cdot 1140.0 \cdot 16.0 \cdot 0.49 \cdot 138 \\ &= 197552. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 311.55) \cdot (12.0 - 1.5) \cdot 0.7 \cdot 138 \\ &= 101229. \text{ Kgf} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg} \\ &= (3.1416/2) \cdot 633.6 \cdot 20.0 \cdot 0.74 \cdot 138 \\ &= 207271. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 633.6 \cdot (20.0 - 1.5) \cdot 0.74 \cdot 138 \\ &= 191725. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (197552 + 101229) = 298781 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw}) \\ &= (68623 + 207271 + 191725 + 0) = 467619 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (S_{pew} + T_{ngw} + S_{inw}) \\ &= (197552 + 191725 + 0) = 389277 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

$$\begin{aligned} \text{Path 1-1} &= 298781 \text{ Kgf, must exceed } W = 131711 \text{ Kgf} & \text{or } W1 &= 154480 \text{ Kgf} \\ \text{Path 2-2} &= 467619 \text{ Kgf, must exceed } W = 131711 \text{ Kgf} & \text{or } W2 &= 13828 \text{ Kgf} \end{aligned}$$



Path 3-3 = 389277 Kgf, must exceed W = 131711 Kgf or W3 = 159947 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 51.5076 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 284.8836 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.931 %

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Input, Nozzle Desc: K1B1.5"

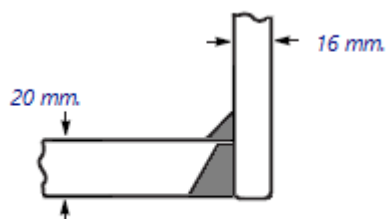
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2000.00	mm.
Design Length of Section	L	4633.3325	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0020	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	207.9620	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K1B1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.500 in.
Actual Thickness Used in Calculation 0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
= 16.8616 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $20.55 \cdot (\exp([23.0 / (138 \cdot 1.0)] - 1))$
= 0.3454 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3519 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.1040 mm.
Parallel to Vessel Wall	Rn+tn+t	53.5520 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1B1.5".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8519 mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 18.3616 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 18.3616 mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.2357 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm.
Wall Thickness per table UG-45	tb3	= 6.0200 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.02, max(18.3616, 3.0)]
= 6.0200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.8519, 6.02)
= 6.0200 mm.

Available Nozzle Neck Thickness = 16.0020 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: K1B1.5"

Intermediate Calc. for nozzle/shell Welds	Tmin	14.5020 mm.
---	------	-------------

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------

PV Elite 28 U2 Licensee: SL User
FileName : 401-V-32003 - 7%
Nozzle Calcs.: K1B_1.5"

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HEXAGON

Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6145 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.5766 mm.

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Input, Nozzle Desc: K1A1.5"

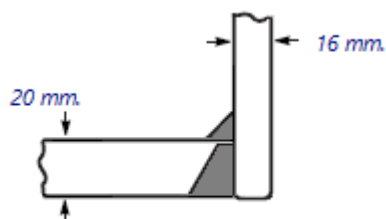
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2000.00	mm.
Design Length of Section	L	4633.3325	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		800.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0020	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	207.9620	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K1A1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.500 in.
Actual Thickness Used in Calculation 0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
= 16.8616 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $20.55 \cdot (\exp([23.0 / (138 \cdot 1.0)] - 1))$
= 0.3454 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3519 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	107.1040 mm.
Parallel to Vessel Wall	Rn+tn+t	53.5520 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1A1.5".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8519 mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 18.3616 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 18.3616 mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.2357 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm.
Wall Thickness per table UG-45	tb3	= 6.0200 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.02, max(18.3616, 3.0)]
= 6.0200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.8519, 6.02)
= 6.0200 mm.

Available Nozzle Neck Thickness = 16.0020 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: K1A1.5"

Intermediate Calc. for nozzle/shell Welds	Tmin	14.5020 mm.
---	------	-------------

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------

PV Elite 28 U2 Licensee: SL User
FileName : 401-V-32003 - 7%
Nozzle Calcs.: K1A_1.5"

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HEXAGON

Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6145 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.5766 mm.

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Input, Nozzle Desc: K2A3"

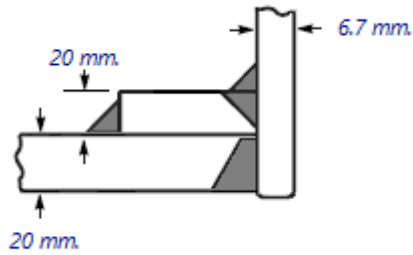
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2000.00	mm.
Design Length of Section	L	4633.3325	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		800.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	303.0760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	152.4000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	16.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		31.7500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2A3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.975 in.
Actual Thickness Used in Calculation	0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
= 16.8616 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.0 \cdot 39.28 / (118 \cdot 1.0 - 0.6 \cdot 23.0)$
= 0.7748 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4709 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	157.1300 mm.
Parallel to Vessel Wall, opening length	d	78.5650 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.9188 mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 138.0)$
= 0.855

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 138.0)$
= 0.855

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.855, 1.0)$
= 0.855

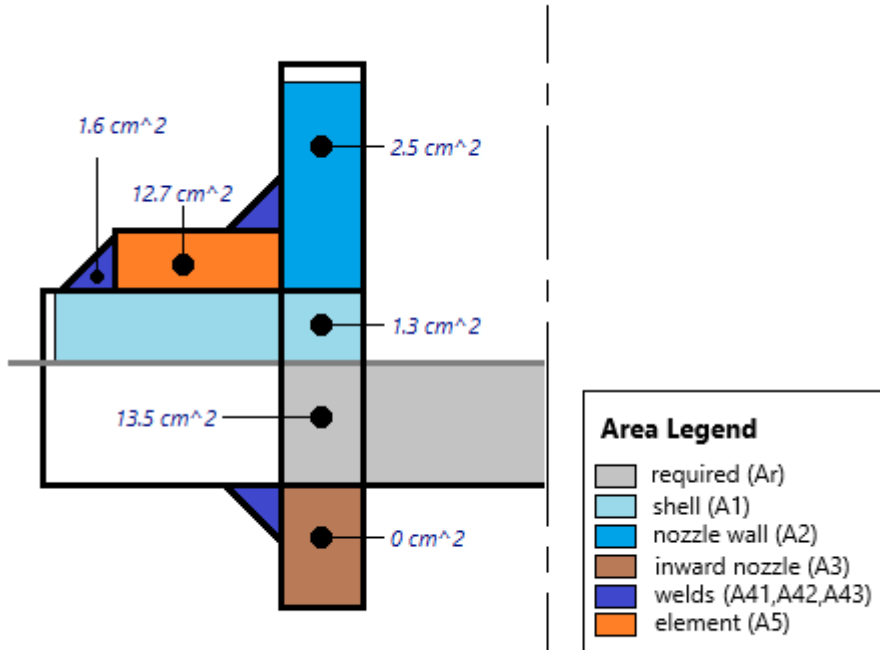


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	13.500	3.649	NA
Area in Shell	A1	1.263	7.233	NA
Area in Nozzle Wall	A2	2.473	2.644	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.556	1.556	NA
Area in Element	A5	12.700	12.700	NA
TOTAL AREA AVAILABLE	Atot	17.992	24.132	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 129.9419 20.0000 mm.
 Based on given Pad Diameter: 152.4000 12.9266 mm.
 Based on the Estimated Diameter Limit: 155.5425 12.3170 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (78.565 \cdot 16.8616 \cdot 1.0 + 2 \cdot 5.1675 \cdot 16.8616 \cdot 1.0 \cdot (1 - 0.86))$
 $= 13.500 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 78.565(1.0 \cdot 18.5 - 1.0 \cdot 16.862) - 2 \cdot 5.168$
 $(1.0 \cdot 18.5 - 1.0 \cdot 16.8616) \cdot (1 - 0.855)$
 $= 1.263 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.92) \cdot (5.17 - 0.77) \cdot 0.8551$



$$= 2.473 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (1.) • 0.86 + (0.0) • 0.86 + 70.0867² • 1.0
= 1.556 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (152.4 - 88.9) • 20.0 • 1.0
= 12.700 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.2748 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 18.3616 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 18.3616 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.2357 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(18.3616, 3.0)]
= 6.3000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.2748, 6.3)
= 6.3000 mm.

Available Nozzle Neck Thickness = 6.6675 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -38 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -38 °C

Weld Size Calculations, Description: K2A3"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.1675 mm.
---	------	------------



Intermediate Calc. for pad/shell Welds TminPad 18.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.6173 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$9.2500 = 0.5 * T_{minPad}$	$11.3120 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (13.4999 - 1.2626 + 2 \cdot 5.1675 \cdot 0.8551 \cdot (1.0 \cdot 18.5 - 16.8616)) \cdot 138)$
 $= 17423.39 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (2.4729 + 12.7 + 1.5559 - 0.0 \cdot 0.86) \cdot 138$
 $= 23540.05 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (2.4729 + 0.0 + 0.8551 + (1.6349)) \cdot 138$
 $= 6983.52 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (2.4729 + 0.0 + 1.5559 + 12.7 + (1.6349)) \cdot 138$
 $= 25840.58 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 8233. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 152.4 \cdot 16.0 \cdot 0.49 \cdot 138$
 $= 26410. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 41.8662) \cdot (6.6675 - 1.5) \cdot 0.7 \cdot 118$
 $= 5725. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 88.9 \cdot 20.0 \cdot 0.74 \cdot 138$
 $= 29082. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 88.9 \cdot (20.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 26901. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NW}) = (26410 + 5725) = 32134 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$
 $= (8233 + 29082 + 26901 + 0) = 64216 \text{ Kgf}$



Failure Path [Path 3-3]:
= (Spew + Tngw + Sinw)
= (26410 + 26901 + 0) = 53310 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 32134 Kgf, must exceed W = 17423 Kgf	or W1 = 23540 Kgf
Path 2-2 = 64215 Kgf, must exceed W = 17423 Kgf	or W2 = 6983 Kgf
Path 3-3 = 53310 Kgf, must exceed W = 17423 Kgf	or W3 = 25840 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.9884 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.0645 mm.

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Input, Nozzle Desc: K2B2"

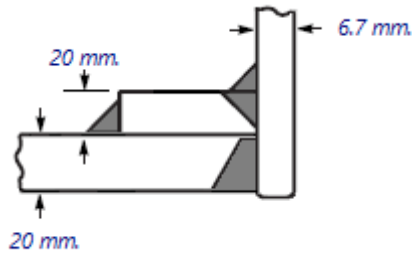
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2000.00	mm.
Design Length of Section	L	4633.3325	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	303.0760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	152.4000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	16.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		31.7500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.975 in.
Actual Thickness Used in Calculation	0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 16.8616 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 39.28 / (118 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 0.7748 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4709 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	157.1300 mm.
Parallel to Vessel Wall, opening length	d	78.5650 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.9188 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

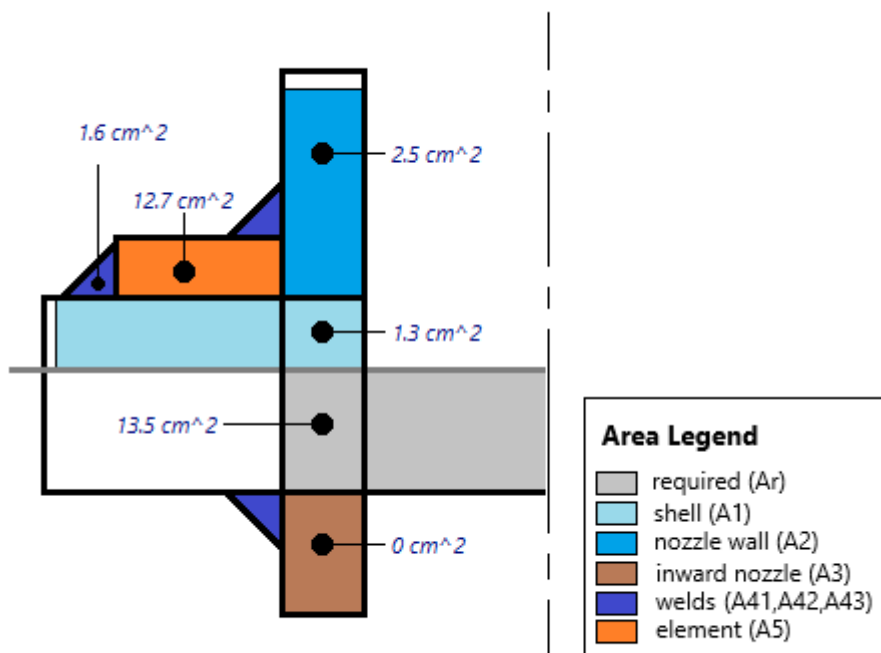


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	13.500	3.649	NA
Area in Shell	A1	1.263	7.233	NA
Area in Nozzle Wall	A2	2.473	2.644	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.556	1.556	NA
Area in Element	A5	12.700	12.700	NA
TOTAL AREA AVAILABLE	Atot	17.992	24.132	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 129.9419 20.0000 mm.
 Based on given Pad Diameter: 152.4000 12.9266 mm.
 Based on the Estimated Diameter Limit: 155.5425 12.3170 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (78.565 \cdot 16.8616 \cdot 1.0 + 2 \cdot 5.1675 \cdot 16.8616 \cdot 1.0 \cdot (1 - 0.86))$
 $= 13.500 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 78.565(1.0 \cdot 18.5 - 1.0 \cdot 16.862) - 2 \cdot 5.168$
 $(1.0 \cdot 18.5 - 1.0 \cdot 16.8616) \cdot (1 - 0.855)$
 $= 1.263 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.92) \cdot (5.17 - 0.77) \cdot 0.8551$



$$= 2.473 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (1.) • 0.86 + (0.0) • 0.86 + 70.0867² • 1.0
= 1.556 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (152.4 - 88.9) • 20.0 • 1.0
= 12.700 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.2748 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 18.3616 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 18.3616 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.2357 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(18.3616, 3.0)]
= 6.3000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.2748, 6.3)
= 6.3000 mm.

Available Nozzle Neck Thickness = 6.6675 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -38 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -38 °C

Weld Size Calculations, Description: K2B2"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.1675 mm.
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Intermediate Calc. for pad/shell Welds TminPad 18.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.6173 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$9.2500 = 0.5 * T_{minPad}$	$11.3120 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (13.4999 - 1.2626 + 2 \cdot 5.1675 \cdot 0.8551 \cdot (1.0 \cdot 18.5 - 16.8616)) \cdot 138)$
 $= 17423.39 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f r_2) \cdot S_v$
 $= (2.4729 + 12.7 + 1.5559 - 0.0 \cdot 0.86) \cdot 138$
 $= 23540.05 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v$
 $= (2.4729 + 0.0 + 0.8551 + (1.6349)) \cdot 138$
 $= 6983.52 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$
 $= (2.4729 + 0.0 + 1.5559 + 12.7 + (1.6349)) \cdot 138$
 $= 25840.58 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 8233. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 152.4 \cdot 16.0 \cdot 0.49 \cdot 138$
 $= 26410. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 41.8662) \cdot (6.6675 - 1.5) \cdot 0.7 \cdot 118$
 $= 5725. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 88.9 \cdot 20.0 \cdot 0.74 \cdot 138$
 $= 29082. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 88.9 \cdot (20.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 26901. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NW}) = (26410 + 5725) = 32134 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$
 $= (8233 + 29082 + 26901 + 0) = 64216 \text{ Kgf}$



Failure Path [Path 3-3]:
= (Spew + Tngw + Sinw)
= (26410 + 26901 + 0) = 53310 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 32134	Kgf, must exceed W = 17423 Kgf	or W1 = 23540 Kgf
Path 2-2 = 64215	Kgf, must exceed W = 17423 Kgf	or W2 = 6983 Kgf
Path 3-3 = 53310	Kgf, must exceed W = 17423 Kgf	or W3 = 25840 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.9884 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.0645 mm.

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Input, Nozzle Desc: K2C3"

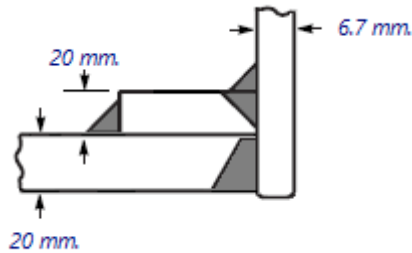
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2000.00	mm.
Design Length of Section	L	4633.3325	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		800.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	303.0760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	152.4000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	16.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		31.7500	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2C3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.975 in.
Actual Thickness Used in Calculation	0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 16.8616 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 39.28 / (118 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 0.7748 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4709 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	157.1300 mm.
Parallel to Vessel Wall, opening length	d	78.5650 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.9188 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

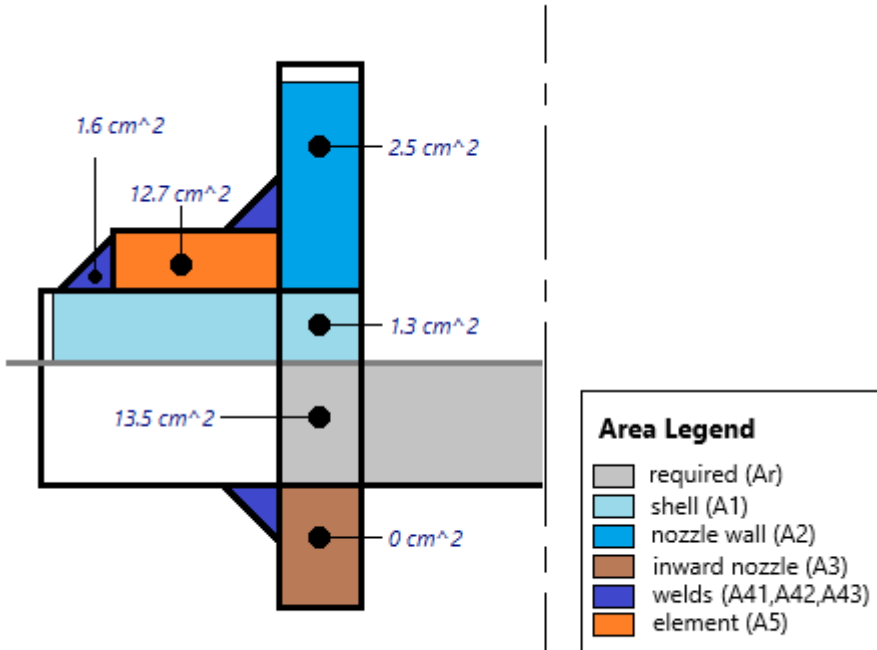


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	13.500	3.649	NA
Area in Shell	A1	1.263	7.233	NA
Area in Nozzle Wall	A2	2.473	2.644	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.556	1.556	NA
Area in Element	A5	12.700	12.700	NA
TOTAL AREA AVAILABLE	Atot	17.992	24.132	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 129.9419 20.0000 mm.
 Based on given Pad Diameter: 152.4000 12.9266 mm.
 Based on the Estimated Diameter Limit: 155.5425 12.3170 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (78.565 \cdot 16.8616 \cdot 1.0 + 2 \cdot 5.1675 \cdot 16.8616 \cdot 1.0 \cdot (1 - 0.86))$
 $= 13.500 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 78.565(1.0 \cdot 18.5 - 1.0 \cdot 16.862) - 2 \cdot 5.168$
 $(1.0 \cdot 18.5 - 1.0 \cdot 16.8616) \cdot (1 - 0.855)$
 $= 1.263 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.92) \cdot (5.17 - 0.77) \cdot 0.8551$



= 2.473 cm²

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (1.) • 0.86 + (0.0) • 0.86 + 70.0867² • 1.0
= 1.556 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (152.4 - 88.9) • 20.0 • 1.0
= 12.700 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures ta = 2.2748 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= ta
= 2.2748 mm.

Available Nozzle Neck Thickness = 6.6675 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Gov. MDMT of the Nozzle : -104 °C

Gov. MDMT of the Reinforcement Pad : -38 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -38 °C

Weld Size Calculations, Description: K2C3"

Intermediate Calc. for nozzle/shell Welds Tmin 5.1675 mm.

Intermediate Calc. for pad/shell Welds TminPad 18.5000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness

Nozzle Weld 3.6173 = 0.7 * tmin. 7.0700 = 0.7 * Wo mm.

Pad Weld 9.2500 = 0.5*TminPad 11.3120 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)

= max(0, (13.4999 - 1.2626 + 2 • 5.1675 • 0.8551 •



$$(1.0 \cdot 18.5 - 16.8616) \cdot 138) \\ = 17423.39 \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv \\ = (2.4729 + 12.7 + 1.5559 - 0.0 \cdot 0.86) \cdot 138 \\ = 23540.05 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ = (2.4729 + 0.0 + 0.8551 + (1.6349)) \cdot 138 \\ = 6983.52 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ = (2.4729 + 0.0 + 1.5559 + 12.7 + (1.6349)) \cdot 138 \\ = 25840.58 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ = (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118 \\ = 8233. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ = (3.1416/2.0) \cdot 152.4 \cdot 16.0 \cdot 0.49 \cdot 138 \\ = 26410. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ = (3.1416 \cdot 41.8662) \cdot (6.6675 - 1.5) \cdot 0.7 \cdot 118 \\ = 5725. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ = (3.1416/2) \cdot 88.9 \cdot 20.0 \cdot 0.74 \cdot 138 \\ = 29082. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ = (3.1416/2.0) \cdot 88.9 \cdot (20.0 - 1.5) \cdot 0.74 \cdot 138 \\ = 26901. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (26410 + 5725) = 32134 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw) \\ = (8233 + 29082 + 26901 + 0) = 64216 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw) \\ = (26410 + 26901 + 0) = 53310 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 32134	Kgf, must exceed W = 17423 Kgf	or W1 = 23540 Kgf
Path 2-2 = 64215	Kgf, must exceed W = 17423 Kgf	or W2 = 6983 Kgf
Path 3-3 = 53310	Kgf, must exceed W = 17423 Kgf	or W3 = 25840 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

PV Elite 28 U2 Licensee: SL User
FileName : 401-V-32003 - 7%
Nozzle Calcs.: K2C_3"

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2:49pm Aug 11, 2026



HEXAGON

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.9884 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.0645 mm.

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Input, Nozzle Desc: K2D3"

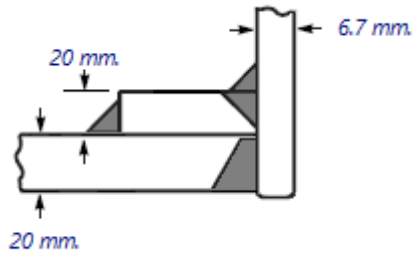
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2000.00	mm.
Design Length of Section	L	4633.3325	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	303.0760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	152.4000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	16.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		31.7500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2D3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.975 in.
Actual Thickness Used in Calculation	0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
= 16.8616 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.0 \cdot 39.28 / (118 \cdot 1.0 - 0.6 \cdot 23.0)$
= 0.7748 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4709 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	157.1300 mm.
Parallel to Vessel Wall, opening length	d	78.5650 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.9188 mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 138.0)$
= 0.855

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 138.0)$
= 0.855

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.855, 1.0)$
= 0.855

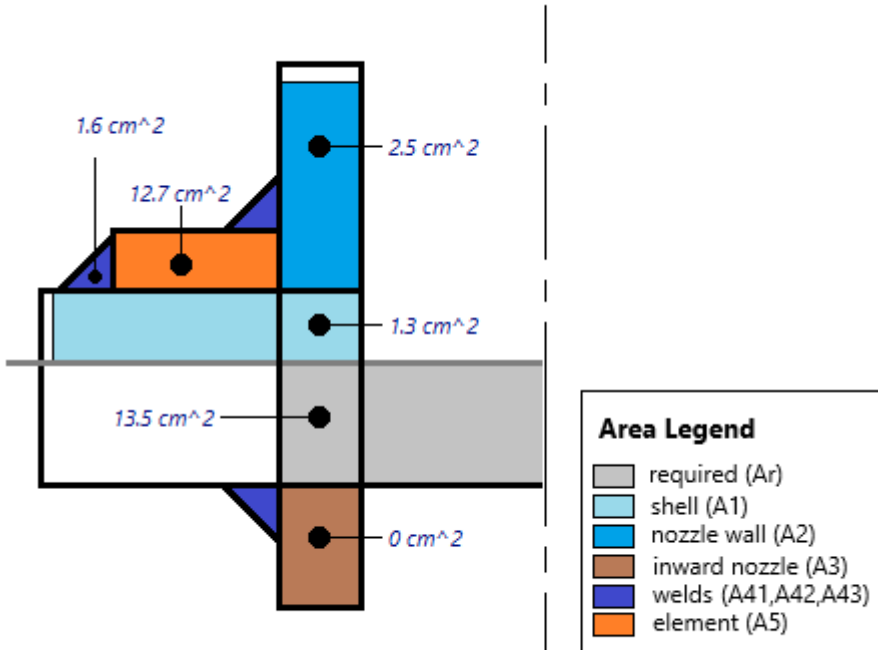


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	13.500	3.649	NA
Area in Shell	A1	1.263	7.233	NA
Area in Nozzle Wall	A2	2.473	2.644	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.556	1.556	NA
Area in Element	A5	12.700	12.700	NA
TOTAL AREA AVAILABLE	Atot	17.992	24.132	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 129.9419 20.0000 mm.
 Based on given Pad Diameter: 152.4000 12.9266 mm.
 Based on the Estimated Diameter Limit: 155.5425 12.3170 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (78.565 \cdot 16.8616 \cdot 1.0 + 2 \cdot 5.1675 \cdot 16.8616 \cdot 1.0 \cdot (1 - 0.86))$
 $= 13.500 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 78.565(1.0 \cdot 18.5 - 1.0 \cdot 16.862) - 2 \cdot 5.168$
 $(1.0 \cdot 18.5 - 1.0 \cdot 16.8616) \cdot (1 - 0.855)$
 $= 1.263 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.92) \cdot (5.17 - 0.77) \cdot 0.8551$



= 2.473 cm²

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (1.) • 0.86 + (0.0) • 0.86 + 70.0867² • 1.0
= 1.556 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (152.4 - 88.9) • 20.0 • 1.0
= 12.700 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.2748 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 18.3616 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 18.3616 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.2357 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(18.3616, 3.0)]
= 6.3000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.2748, 6.3)
= 6.3000 mm.

Available Nozzle Neck Thickness = 6.6675 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -38 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -38 °C

Weld Size Calculations, Description: K2D3"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.1675 mm.
---	------	------------



Intermediate Calc. for pad/shell Welds TminPad 18.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.6173 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$9.2500 = 0.5 * T_{minPad}$	$11.3120 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (13.4999 - 1.2626 + 2 \cdot 5.1675 \cdot 0.8551 \cdot (1.0 \cdot 18.5 - 16.8616)) \cdot 138)$
 $= 17423.39 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f r_2) \cdot S_v$
 $= (2.4729 + 12.7 + 1.5559 - 0.0 \cdot 0.86) \cdot 138$
 $= 23540.05 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v$
 $= (2.4729 + 0.0 + 0.8551 + (1.6349)) \cdot 138$
 $= 6983.52 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$
 $= (2.4729 + 0.0 + 1.5559 + 12.7 + (1.6349)) \cdot 138$
 $= 25840.58 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 8233. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 152.4 \cdot 16.0 \cdot 0.49 \cdot 138$
 $= 26410. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 41.8662) \cdot (6.6675 - 1.5) \cdot 0.7 \cdot 118$
 $= 5725. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 88.9 \cdot 20.0 \cdot 0.74 \cdot 138$
 $= 29082. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 88.9 \cdot (20.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 26901. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NW}) = (26410 + 5725) = 32134 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$
 $= (8233 + 29082 + 26901 + 0) = 64216 \text{ Kg}$



Failure Path [Path 3-3]:
= (Spew + Tngw + Sinw)
= (26410 + 26901 + 0) = 53310 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 32134 Kgf, must exceed W = 17423 Kgf	or W1 = 23540 Kgf
Path 2-2 = 64215 Kgf, must exceed W = 17423 Kgf	or W2 = 6983 Kgf
Path 3-3 = 53310 Kgf, must exceed W = 17423 Kgf	or W3 = 25840 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.9884 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.0645 mm.

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Input, Nozzle Desc: K2E3"

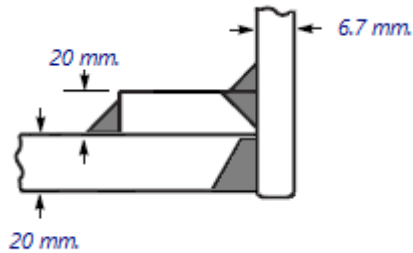
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2000.00	mm.
Design Length of Section	L	4633.3325	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		800.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		110.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	303.0760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	152.4000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	16.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		31.7500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2E3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.975 in.
Actual Thickness Used in Calculation	0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 16.8616$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 39.28 / (118 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 0.7748$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4709 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	157.1300 mm.
Parallel to Vessel Wall, opening length	d	78.5650 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.9188 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

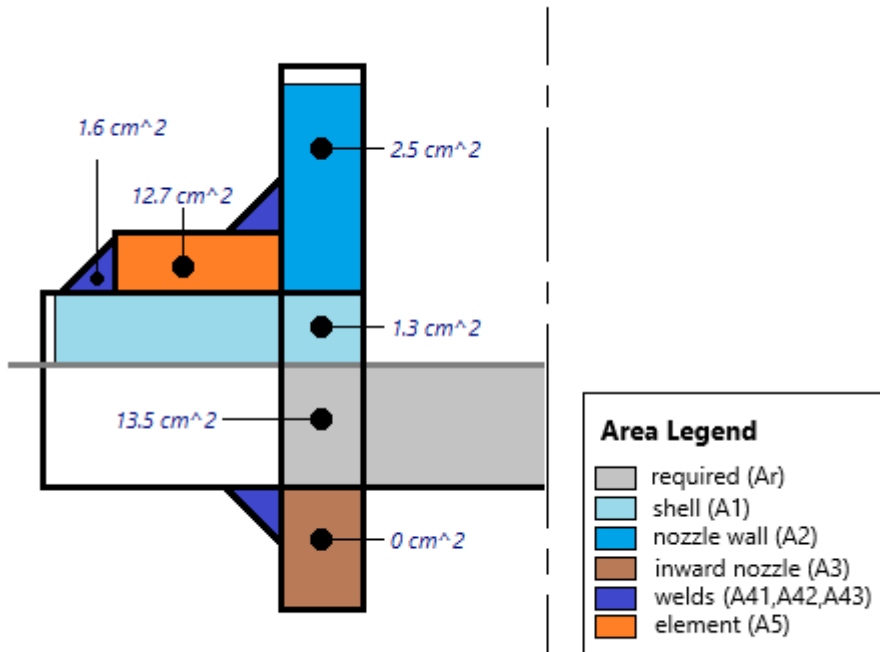


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	13.500	3.649	NA
Area in Shell	A1	1.263	7.233	NA
Area in Nozzle Wall	A2	2.473	2.644	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.556	1.556	NA
Area in Element	A5	12.700	12.700	NA
TOTAL AREA AVAILABLE	Atot	17.992	24.132	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 129.9419 20.0000 mm.
 Based on given Pad Diameter: 152.4000 12.9266 mm.
 Based on the Estimated Diameter Limit: 155.5425 12.3170 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (78.565 \cdot 16.8616 \cdot 1.0 + 2 \cdot 5.1675 \cdot 16.8616 \cdot 1.0 \cdot (1 - 0.86))$
 $= 13.500 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 78.565(1.0 \cdot 18.5 - 1.0 \cdot 16.862) - 2 \cdot 5.168$
 $(1.0 \cdot 18.5 - 1.0 \cdot 16.8616) \cdot (1 - 0.855)$
 $= 1.263 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.92) \cdot (5.17 - 0.77) \cdot 0.8551$



$$= 2.473 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (1.) • 0.86 + (0.0) • 0.86 + 70.0867² • 1.0
= 1.556 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (152.4 - 88.9) • 20.0 • 1.0
= 12.700 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.2748 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 18.3616 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 18.3616 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.2357 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(18.3616, 3.0)]
= 6.3000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.2748, 6.3)
= 6.3000 mm.

Available Nozzle Neck Thickness = 6.6675 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -38 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -38 °C

Weld Size Calculations, Description: K2E3"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.1675 mm.
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Intermediate Calc. for pad/shell Welds TminPad 18.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.6173 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$9.2500 = 0.5 * T_{minPad}$	$11.3120 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (13.4999 - 1.2626 + 2 \cdot 5.1675 \cdot 0.8551 \cdot (1.0 \cdot 18.5 - 16.8616)) \cdot 138)$
 $= 17423.39 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f r_2) \cdot S_v$
 $= (2.4729 + 12.7 + 1.5559 - 0.0 \cdot 0.86) \cdot 138$
 $= 23540.05 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v$
 $= (2.4729 + 0.0 + 0.8551 + (1.6349)) \cdot 138$
 $= 6983.52 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$
 $= (2.4729 + 0.0 + 1.5559 + 12.7 + (1.6349)) \cdot 138$
 $= 25840.58 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 8233. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 152.4 \cdot 16.0 \cdot 0.49 \cdot 138$
 $= 26410. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 41.8662) \cdot (6.6675 - 1.5) \cdot 0.7 \cdot 118$
 $= 5725. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 88.9 \cdot 20.0 \cdot 0.74 \cdot 138$
 $= 29082. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 88.9 \cdot (20.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 26901. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NW}) = (26410 + 5725) = 32134 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$
 $= (8233 + 29082 + 26901 + 0) = 64216 \text{ Kg}$



Failure Path [Path 3-3]:
= (Spew + Tngw + Sinw)
= (26410 + 26901 + 0) = 53310 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 32134 Kgf, must exceed W = 17423 Kgf	or W1 = 23540 Kgf
Path 2-2 = 64215 Kgf, must exceed W = 17423 Kgf	or W2 = 6983 Kgf
Path 3-3 = 53310 Kgf, must exceed W = 17423 Kgf	or W3 = 25840 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.9884 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.0645 mm.

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Input, Nozzle Desc: K2F3"

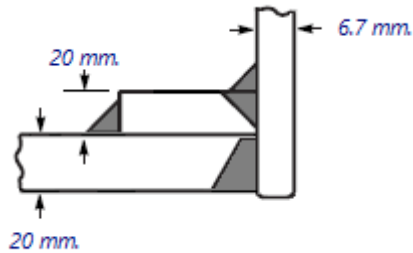
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2000.00	mm.
Design Length of Section	L	4633.3325	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		110.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	303.0760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	152.4000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	16.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		31.7500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2F3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.975 in.
Actual Thickness Used in Calculation	0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
= 16.8616 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.0 \cdot 39.28 / (118 \cdot 1.0 - 0.6 \cdot 23.0)$
= 0.7748 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4709 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	157.1300 mm.
Parallel to Vessel Wall, opening length	d	78.5650 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.9188 mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 138.0)$
= 0.855

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 138.0)$
= 0.855

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.855, 1.0)$
= 0.855

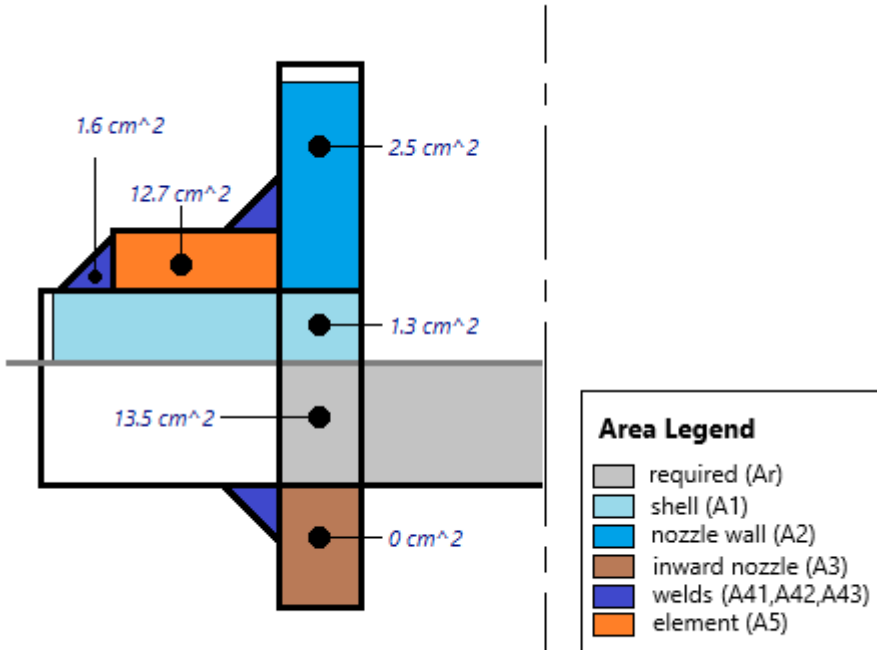


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	13.500	3.649	NA
Area in Shell	A1	1.263	7.233	NA
Area in Nozzle Wall	A2	2.473	2.644	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.556	1.556	NA
Area in Element	A5	12.700	12.700	NA
TOTAL AREA AVAILABLE	Atot	17.992	24.132	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 129.9419 20.0000 mm.
 Based on given Pad Diameter: 152.4000 12.9266 mm.
 Based on the Estimated Diameter Limit: 155.5425 12.3170 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (78.565 \cdot 16.8616 \cdot 1.0 + 2 \cdot 5.1675 \cdot 16.8616 \cdot 1.0 \cdot (1 - 0.86))$
 $= 13.500 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 78.565(1.0 \cdot 18.5 - 1.0 \cdot 16.862) - 2 \cdot 5.168$
 $(1.0 \cdot 18.5 - 1.0 \cdot 16.8616) \cdot (1 - 0.855)$
 $= 1.263 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.92) \cdot (5.17 - 0.77) \cdot 0.8551$



= 2.473 cm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
 = (1.) • 0.86 + (0.0) • 0.86 + 70.0867² • 1.0
 = 1.556 cm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (152.4 - 88.9) • 20.0 • 1.0
 = 12.700 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.2748 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 18.3616 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 18.3616 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.2357 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[6.3, max(18.3616, 3.0)]
 = 6.3000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(2.2748, 6.3)
 = 6.3000 mm.

Available Nozzle Neck Thickness = 6.6675 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -38 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -38 °C

Weld Size Calculations, Description: K2F3"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.1675 mm.
---	------	------------



Intermediate Calc. for pad/shell Welds TminPad 18.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.6173 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$9.2500 = 0.5 * T_{minPad}$	$11.3120 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (13.4999 - 1.2626 + 2 \cdot 5.1675 \cdot 0.8551 \cdot (1.0 \cdot 18.5 - 16.8616)) \cdot 138)$
 $= 17423.39 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f r_2) \cdot S_v$
 $= (2.4729 + 12.7 + 1.5559 - 0.0 \cdot 0.86) \cdot 138$
 $= 23540.05 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v$
 $= (2.4729 + 0.0 + 0.8551 + (1.6349)) \cdot 138$
 $= 6983.52 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$
 $= (2.4729 + 0.0 + 1.5559 + 12.7 + (1.6349)) \cdot 138$
 $= 25840.58 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 8233. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 152.4 \cdot 16.0 \cdot 0.49 \cdot 138$
 $= 26410. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 41.8662) \cdot (6.6675 - 1.5) \cdot 0.7 \cdot 118$
 $= 5725. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 88.9 \cdot 20.0 \cdot 0.74 \cdot 138$
 $= 29082. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 88.9 \cdot (20.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 26901. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NW}) = (26410 + 5725) = 32134 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$
 $= (8233 + 29082 + 26901 + 0) = 64216 \text{ Kg}$



Failure Path [Path 3-3]:
= (Spew + Tngw + Sinw)
= (26410 + 26901 + 0) = 53310 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 32134	Kgf, must exceed W = 17423 Kgf	or W1 = 23540 Kgf
Path 2-2 = 64215	Kgf, must exceed W = 17423 Kgf	or W2 = 6983 Kgf
Path 3-3 = 53310	Kgf, must exceed W = 17423 Kgf	or W3 = 25840 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.9884 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.0645 mm.

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Input, Nozzle Desc: K3A3"

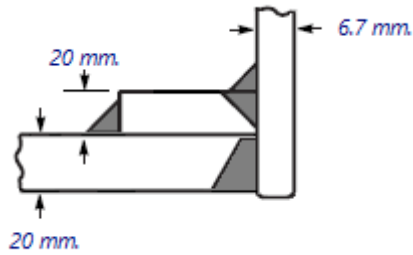
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2000.00	mm.
Design Length of Section	L	4633.3325	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	700.0000	mm.
Distance from Bottom/Left Tangent		800.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		236.66	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	303.0760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	152.4000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	16.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		31.7500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K3A3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.975 in.
Actual Thickness Used in Calculation	0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 16.8616 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 39.28 / (118 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 0.7748 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4709 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	218.3884 mm.
Parallel to Vessel Wall, opening length	d	109.1942 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.9188 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

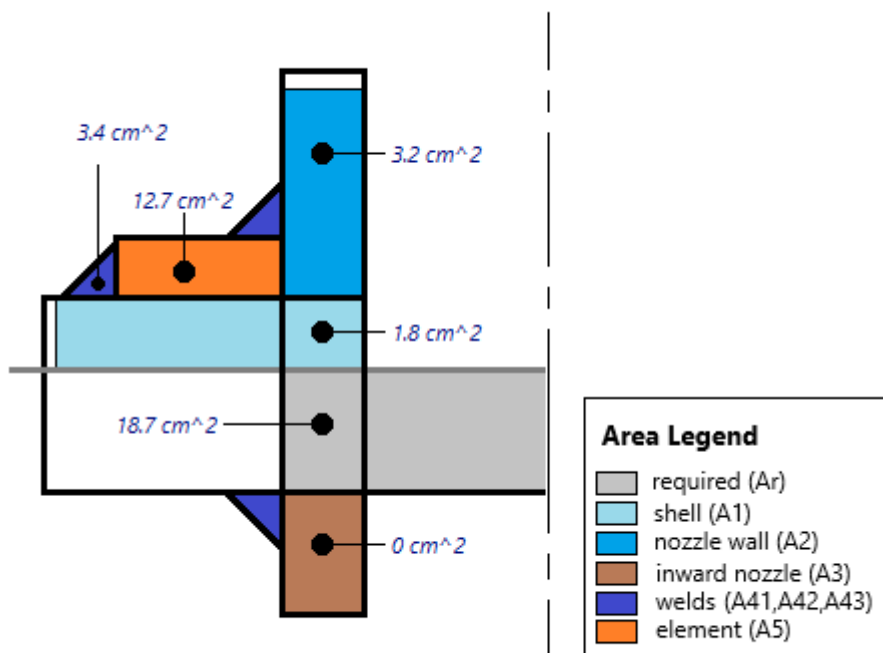


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	18.664	5.045	NA
Area in Shell	A1	1.764	10.107	NA
Area in Nozzle Wall	A2	3.246	3.471	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.414	3.414	NA
Area in Element	A5	12.700	12.700	NA
TOTAL AREA AVAILABLE	Atot	21.124	29.691	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 46.01 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 140.1004 20.0000 mm.
 Based on given Pad Diameter: 152.4000 16.1261 mm.
 Based on the Estimated Diameter Limit: 216.8009 10.9822 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (109.1942 \cdot 16.8616 \cdot 1.0 + 2 \cdot 5.1675 \cdot 16.8616 \cdot 1.0 \cdot (1 - 0.86))$
 $= 18.664 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 109.194(1.0 \cdot 18.5 - 1.0 \cdot 16.862) - 2 \cdot 5.168$
 $(1.0 \cdot 18.5 - 1.0 \cdot 16.8616) \cdot (1 - 0.855)$
 $= 1.764 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 32.92) \cdot (5.17 - 0.77) \cdot 0.8551 / \sin(49.6)$



= 3.246 cm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
 = (1.) • 0.86 + (0.0) • 0.86 + 255.8878² • 1.0
 = 3.414 cm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (187.0584 - 123.5584) • 20.0 • 1.0
 = 12.700 cm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -38 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -38 °C

Weld Size Calculations, Description: K3A3"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.1675 mm.
Intermediate Calc. for pad/shell Welds	TminPad	18.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	3.6173 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.
Pad Weld	9.2500 = 0.5*TminPad	11.3120 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
 = max(0, (18.6645 - 1.7645 + 2 • 5.1675 • 0.8551 •
 (1.0 • 18.5 - 16.8616))138)
 = 23984.60 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 = (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv



$$= (3.246 + 12.7 + 3.414 - 0.0 \cdot 0.86) \cdot 138$$
$$= 27242.35 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$
$$= (3.246 + 0.0 + 0.8551 + (1.6349)) \cdot 138$$
$$= 8071.32 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$
$$= (3.246 + 0.0 + 3.414 + 12.7 + (1.6349)) \cdot 138$$
$$= 29542.88 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$
$$= (3.1416/2.0) \cdot 123.5584 \cdot 10.0 \cdot 0.49 \cdot 118$$
$$= 11443. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$
$$= (3.1416/2.0) \cdot 152.4 \cdot 16.0 \cdot 0.49 \cdot 138$$
$$= 26410. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$
$$= (3.1416 \cdot 58.1881) \cdot (6.6675 - 1.5) \cdot 0.7 \cdot 118$$
$$= 7956. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$
$$= (3.1416/2) \cdot 123.5584 \cdot 20.0 \cdot 0.74 \cdot 138$$
$$= 40420. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$
$$= (3.1416/2.0) \cdot 123.5584 \cdot (20.0 - 1.5) \cdot 0.74 \cdot 138$$
$$= 37388. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (26410 + 7956) = 34366 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (11443 + 40420 + 37388 + 0) = 89251 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$
$$= (26410 + 37388 + 0) = 63798 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 34365	Kgf, must exceed W = 23984 Kgf	or W1 = 27242 Kgf
Path 2-2 = 89251	Kgf, must exceed W = 23984 Kgf	or W2 = 8071 Kgf
Path 3-3 = 63797	Kgf, must exceed W = 23984 Kgf	or W3 = 29542 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Nozzle is O.K. for the External Pressure: 1 bars

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K3A3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.975 in.
 Actual Thickness Used in Calculation 0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 16.8616 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 39.28 / (118 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 0.7748 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4709 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 157.1300 mm.
 Parallel to Vessel Wall, opening length d 78.5650 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 32.9188 mm.

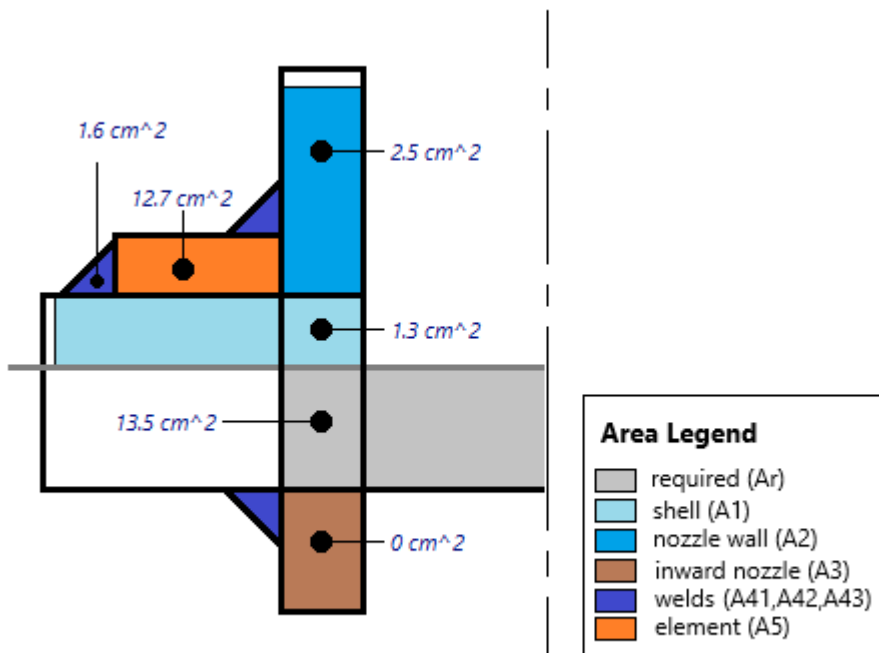


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	13.500	3.649	NA
Area in Shell	A1	1.263	7.233	NA
Area in Nozzle Wall	A2	2.473	2.644	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.556	1.556	NA
Area in Element	A5	12.700	12.700	NA
TOTAL AREA AVAILABLE	Atot	17.992	24.132	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.



The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	129.9419	20.0000 mm.
Based on given Pad Diameter:	152.4000	12.9266 mm.
Based on the Estimated Diameter Limit:	155.5425	12.3170 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (78.565 \cdot 16.8616 \cdot 1.0 + 2 \cdot 5.1675 \cdot 16.8616 \cdot 1.0 \cdot (1 - 0.86))$
 $= 13.500 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 78.565(1.0 \cdot 18.5 - 1.0 \cdot 16.862) - 2 \cdot 5.168$
 $(1.0 \cdot 18.5 - 1.0 \cdot 16.8616) \cdot (1 - 0.855)$
 $= 1.263 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.92) \cdot (5.17 - 0.77) \cdot 0.8551$
 $= 2.473 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$
 $= (1.) \cdot 0.86 + (0.0) \cdot 0.86 + 70.0867^2 \cdot 1.0$
 $= 1.556 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (152.4 - 88.9) \cdot 20.0 \cdot 1.0$
 $= 12.700 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.2748 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 18.3616 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 18.3616 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.2357 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(18.3616, 3.0)]$
 $= 6.3000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.2748, 6.3)$
 $= 6.3000 \text{ mm.}$

Available Nozzle Neck Thickness = 6.6675 mm. -> OK

Weld Size Calculations, Description: K3A3"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.1675 mm.
Intermediate Calc. for pad/shell Welds	TminPad	18.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.6173 = 0.7 \cdot tmin.$	$7.0700 = 0.7 \cdot Wo \text{ mm.}$
Pad Weld	$9.2500 = 0.5 \cdot TminPad$	$11.3120 = 0.7 \cdot Wp \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) Sv)$



$$\begin{aligned} &= \max(0, (13.4999 - 1.2626 + 2 \cdot 5.1675 \cdot 0.8551 \cdot \\ &\quad (1.0 \cdot 18.5 - 16.8616)) \cdot 138) \\ &= 17423.39 \text{ Kgf} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

$$\begin{aligned} \text{Weld Load [W1]:} \\ &= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v \\ &= (2.4729 + 12.7 + 1.5559 - 0.0 \cdot 0.86) \cdot 138 \\ &= 23540.05 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Weld Load [W2]:} \\ &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v \\ &= (2.4729 + 0.0 + 0.8551 + (1.6349)) \cdot 138 \\ &= 6983.52 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Weld Load [W3]:} \\ &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S \\ &= (2.4729 + 0.0 + 1.5559 + 12.7 + (1.6349)) \cdot 138 \\ &= 25840.58 \text{ Kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

$$\begin{aligned} \text{Shear, Outward Nozzle Weld [Sonw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118 \\ &= 8233. \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Shear, Pad Element Weld [Spew]:} \\ &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\ &= (3.1416/2.0) \cdot 152.4 \cdot 16.0 \cdot 0.49 \cdot 138 \\ &= 26410. \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Shear, Nozzle Wall [Snw]:} \\ &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 41.8662) \cdot (6.6675 - 1.5) \cdot 0.7 \cdot 118 \\ &= 5725. \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension, Pad Groove Weld [Tpgw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg} \\ &= (3.1416/2) \cdot 88.9 \cdot 20.0 \cdot 0.74 \cdot 138 \\ &= 29082. \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension, Shell Groove Weld [Tngw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 88.9 \cdot (20.0 - 1.5) \cdot 0.74 \cdot 138 \\ &= 26901. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

$$\begin{aligned} \text{Failure Path [Path 1-1]:} \\ &= (S_{PEW} + S_{NW}) = (26410 + 5725) = 32134 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Failure Path [Path 2-2]:} \\ &= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw}) \\ &= (8233 + 29082 + 26901 + 0) = 64216 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Failure Path [Path 3-3]:} \\ &= (S_{pew} + T_{ngw} + S_{inw}) \\ &= (26410 + 26901 + 0) = 53310 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 32134	Kgf, must exceed W = 17423 Kgf	or W1 = 23540 Kgf
Path 2-2 = 64215	Kgf, must exceed W = 17423 Kgf	or W2 = 6983 Kgf
Path 3-3 = 53310	Kgf, must exceed W = 17423 Kgf	or W3 = 25840 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 46.4647 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 377.2873 mm.

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Input, Nozzle Desc: K3B2"

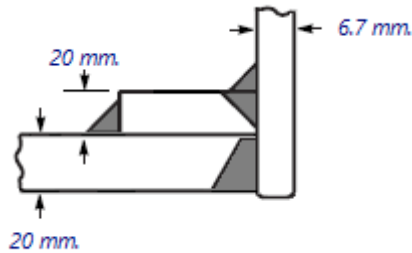
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2000.00	mm.
Design Length of Section	L	4633.3325	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	700.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		236.66	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	303.0760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	152.4000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	16.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		31.7500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K3B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.975 in.
Actual Thickness Used in Calculation	0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 16.8616 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 39.28 / (118 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 0.7748 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4709 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	218.3884 mm.
Parallel to Vessel Wall, opening length	d	109.1942 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.9188 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

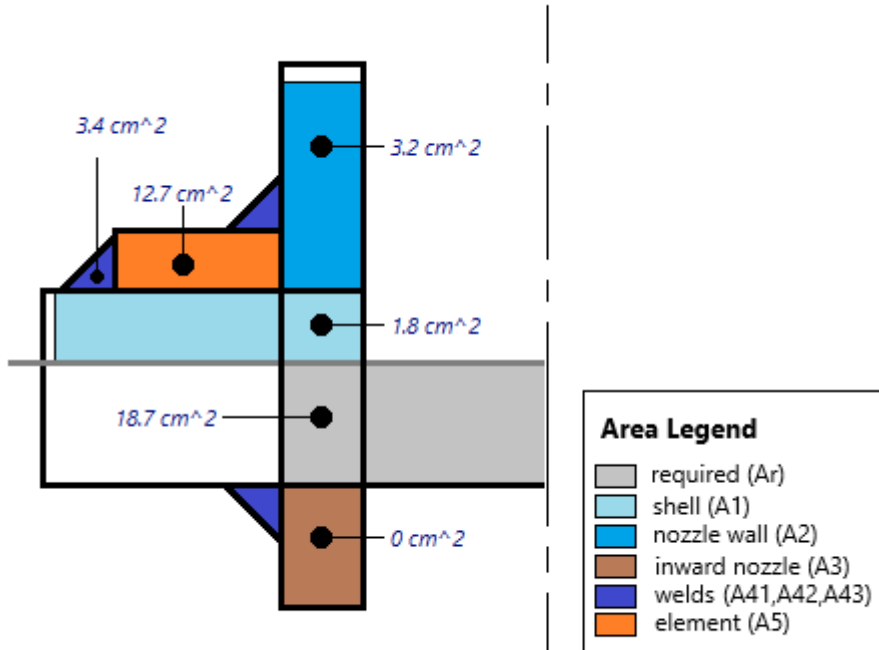


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	18.664	5.045	NA
Area in Shell	A1	1.764	10.107	NA
Area in Nozzle Wall	A2	3.246	3.471	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.414	3.414	NA
Area in Element	A5	12.700	12.700	NA
TOTAL AREA AVAILABLE	Atot	21.124	29.691	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 46.01 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 140.1004 20.0000 mm.
 Based on given Pad Diameter: 152.4000 16.1261 mm.
 Based on the Estimated Diameter Limit: 216.8009 10.9822 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (109.1942 \cdot 16.8616 \cdot 1.0 + 2 \cdot 5.1675 \cdot 16.8616 \cdot 1.0 \cdot (1 - 0.86))$
 $= 18.664 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 109.194(1.0 \cdot 18.5 - 1.0 \cdot 16.862) - 2 \cdot 5.168$
 $(1.0 \cdot 18.5 - 1.0 \cdot 16.8616) \cdot (1 - 0.855)$
 $= 1.764 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 32.92) \cdot (5.17 - 0.77) \cdot 0.8551 / \sin(49.6)$



= 3.246 cm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (1.) • 0.86 + (0.0) • 0.86 + 255.8878² • 1.0
= 3.414 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (187.0584 - 123.5584) • 20.0 • 1.0
= 12.700 cm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Gov. MDMT of the Nozzle	:	-104 °C
Gov. MDMT of the Reinforcement Pad	:	-38 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-38 °C

Weld Size Calculations, Description: K3B2"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.1675 mm.
Intermediate Calc. for pad/shell Welds	TminPad	18.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	3.6173 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.
Pad Weld	9.2500 = 0.5*TminPad	11.3120 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (18.6645 - 1.7645 + 2 • 5.1675 • 0.8551 •
(1.0 • 18.5 - 16.8616))138)
= 23984.60 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv



$$= (3.246 + 12.7 + 3.414 - 0.0 \cdot 0.86) \cdot 138$$
$$= 27242.35 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$
$$= (3.246 + 0.0 + 0.8551 + (1.6349)) \cdot 138$$
$$= 8071.32 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$
$$= (3.246 + 0.0 + 3.414 + 12.7 + (1.6349)) \cdot 138$$
$$= 29542.88 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$
$$= (3.1416/2.0) \cdot 123.5584 \cdot 10.0 \cdot 0.49 \cdot 118$$
$$= 11443. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$
$$= (3.1416/2.0) \cdot 152.4 \cdot 16.0 \cdot 0.49 \cdot 138$$
$$= 26410. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$
$$= (3.1416 \cdot 58.1881) \cdot (6.6675 - 1.5) \cdot 0.7 \cdot 118$$
$$= 7956. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$
$$= (3.1416/2) \cdot 123.5584 \cdot 20.0 \cdot 0.74 \cdot 138$$
$$= 40420. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$
$$= (3.1416/2.0) \cdot 123.5584 \cdot (20.0 - 1.5) \cdot 0.74 \cdot 138$$
$$= 37388. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (26410 + 7956) = 34366 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (11443 + 40420 + 37388 + 0) = 89251 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$
$$= (26410 + 37388 + 0) = 63798 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 34365	Kgf, must exceed W = 23984 Kgf	or W1 = 27242 Kgf
Path 2-2 = 89251	Kgf, must exceed W = 23984 Kgf	or W2 = 8071 Kgf
Path 3-3 = 63797	Kgf, must exceed W = 23984 Kgf	or W3 = 29542 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars

Nozzle is O.K. for the External Pressure: 1 bars

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K3B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.975 in.
 Actual Thickness Used in Calculation 0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 16.8616$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 39.28 / (118 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 0.7748$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4709 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 157.1300 mm.
 Parallel to Vessel Wall, opening length d 78.5650 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 32.9188 mm.

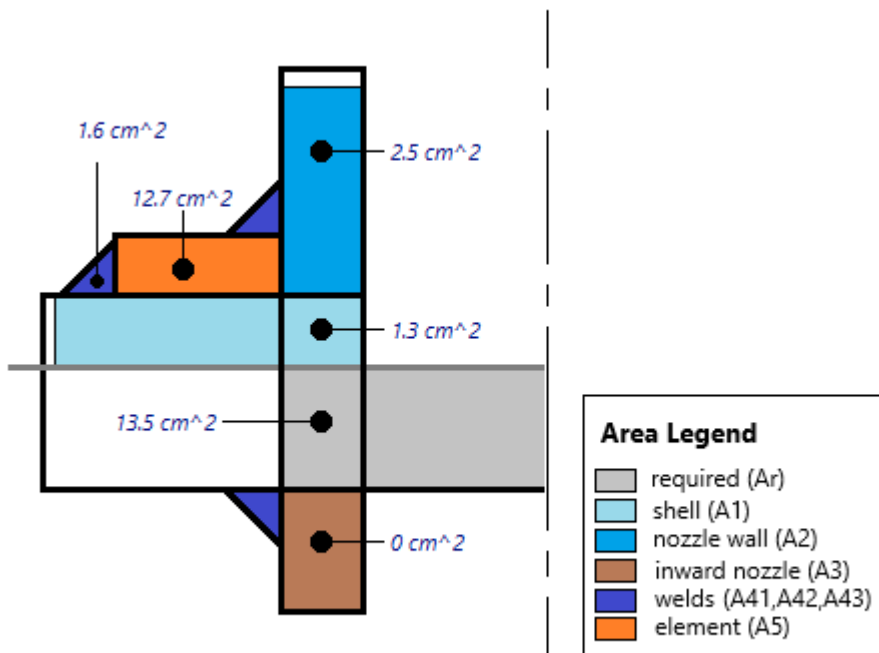


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	13.500	3.649	NA
Area in Shell	A1	1.263	7.233	NA
Area in Nozzle Wall	A2	2.473	2.644	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.556	1.556	NA
Area in Element	A5	12.700	12.700	NA
TOTAL AREA AVAILABLE	Atot	17.992	24.132	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.



The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	129.9419	20.0000 mm.
Based on given Pad Diameter:	152.4000	12.9266 mm.
Based on the Estimated Diameter Limit:	155.5425	12.3170 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (78.565 \cdot 16.8616 \cdot 1.0 + 2 \cdot 5.1675 \cdot 16.8616 \cdot 1.0 \cdot (1 - 0.86))$
 $= 13.500 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 78.565(1.0 \cdot 18.5 - 1.0 \cdot 16.862) - 2 \cdot 5.168$
 $(1.0 \cdot 18.5 - 1.0 \cdot 16.8616) \cdot (1 - 0.855)$
 $= 1.263 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.92) \cdot (5.17 - 0.77) \cdot 0.8551$
 $= 2.473 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$
 $= (1.) \cdot 0.86 + (0.0) \cdot 0.86 + 70.0867^2 \cdot 1.0$
 $= 1.556 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (152.4 - 88.9) \cdot 20.0 \cdot 1.0$
 $= 12.700 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.2748 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 18.3616 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 18.3616 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.2357 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(18.3616, 3.0)]$
 $= 6.3000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.2748, 6.3)$
 $= 6.3000 \text{ mm.}$

Available Nozzle Neck Thickness = 6.6675 mm. -> OK

Weld Size Calculations, Description: K3B2"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.1675 mm.
Intermediate Calc. for pad/shell Welds	TminPad	18.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.6173 = 0.7 \cdot tmin.$	$7.0700 = 0.7 \cdot Wo \text{ mm.}$
Pad Weld	$9.2500 = 0.5 \cdot TminPad$	$11.3120 = 0.7 \cdot Wp \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) Sv)$



$$= \max(0, (13.4999 - 1.2626 + 2 \cdot 5.1675 \cdot 0.8551 \cdot (1.0 \cdot 18.5 - 16.8616)) \cdot 138)$$

$$= 17423.39 \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v$$

$$= (2.4729 + 12.7 + 1.5559 - 0.0 \cdot 0.86) \cdot 138$$

$$= 23540.05 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v$$

$$= (2.4729 + 0.0 + 0.8551 + (1.6349)) \cdot 138$$

$$= 6983.52 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S$$

$$= (2.4729 + 0.0 + 1.5559 + 12.7 + (1.6349)) \cdot 138$$

$$= 25840.58 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 8233. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 152.4 \cdot 16.0 \cdot 0.49 \cdot 138$$

$$= 26410. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 41.8662) \cdot (6.6675 - 1.5) \cdot 0.7 \cdot 118$$

$$= 5725. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 88.9 \cdot 20.0 \cdot 0.74 \cdot 138$$

$$= 29082. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 88.9 \cdot (20.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 26901. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (26410 + 5725) = 32134 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (8233 + 29082 + 26901 + 0) = 64216 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (26410 + 26901 + 0) = 53310 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 32134	Kgf, must exceed W = 17423 Kgf	or W1 = 23540 Kgf
Path 2-2 = 64215	Kgf, must exceed W = 17423 Kgf	or W2 = 6983 Kgf
Path 3-3 = 53310	Kgf, must exceed W = 17423 Kgf	or W3 = 25840 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 25 bars



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 46.4647 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 377.2873 mm.

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Input, Nozzle Desc: N42"

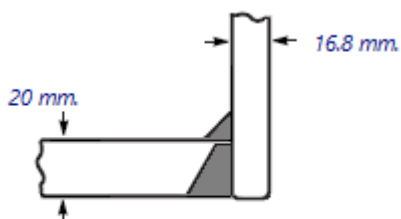
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2000.00	mm.
Design Length of Section	L	4633.3325	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		250.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-333 6	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	309.4260	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $23.0 \cdot 1001.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
= 16.8616 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $26.9 \cdot (\exp([23.0 / (118.0 \cdot 1.0)] - 1))$
= 0.5295 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4601 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	121.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	60.6640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N42".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0295 mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 18.3616 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 18.3616 mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.2357 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm.
Wall Thickness per table UG-45	tb3	= 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(18.3616, 3.0)]
= 6.3000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.0295, 6.3)
= 6.3000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C



Weld Size Calculations, Description: N42"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.8893 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 330.3153 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 60.66 mm.

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Input, Nozzle Desc: N218"

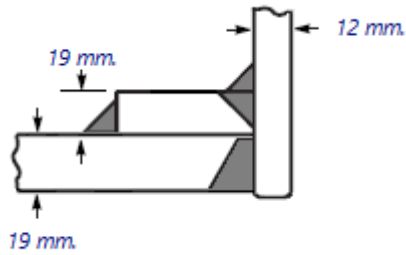
From: 50

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.01	bars
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	19.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	422.7740	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	19.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	800.0000	mm.
Thickness of Pad	te	19.0000	mm.
Weld leg size between Pad and Shell	Wp	16.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	19.0000	mm.
Reinforcing Pad Width		159.4000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N218"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	18.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 23.0 \cdot 2003.0 \cdot 0.8987 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 23.0)$
 $= 15.0273 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 230.1 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 3.8740 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 1.4460 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	920.4000 mm.
Parallel to Vessel Wall, opening length	d	460.2000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		43.7500 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

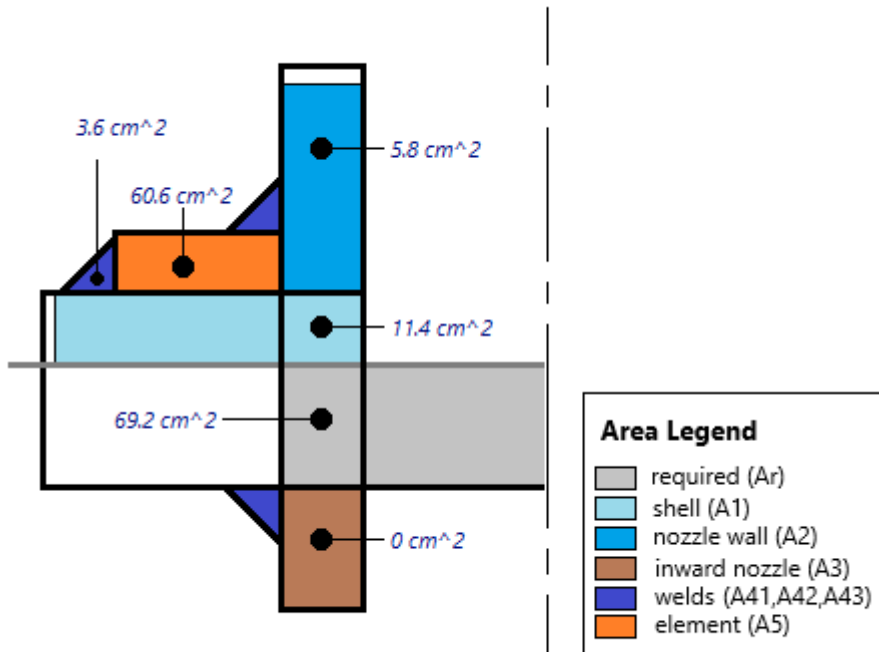


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	69.156	12.018	NA
Area in Shell	A1	11.379	56.499	NA
Area in Nozzle Wall	A2	5.798	7.922	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.560	3.560	NA
Area in Element	A5	60.572	60.572	NA
TOTAL AREA AVAILABLE	Atot	81.309	128.553	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 736.0336 19.0000 mm.
 Based on given Pad Diameter: 800.0000 15.1877 mm.
 Based on Shell or Nozzle Thickness: 884.6866 12.0000 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (460.2 \cdot 15.0273 \cdot 1.0 + 2 \cdot 10.5 \cdot 15.0273 \cdot 1.0 \cdot (1 - 1.0))$
 $= 69.156 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 460.2(1.0 \cdot 17.5 - 1.0 \cdot 15.027) - 2 \cdot 10.5$
 $(1.0 \cdot 17.5 - 1.0 \cdot 15.0273) \cdot (1 - 1.0)$
 $= 11.379 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 43.75) \cdot (10.5 - 3.87) \cdot 1.0$



= 5.798 cm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 16.0^2 \cdot 1.0$
= 3.560 cm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
= $(800.0 - 481.2) \cdot 19.0 \cdot 1.0$
= 60.572 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.3740 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 18.1876 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 18.1876 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.2340 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 9.8312 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[9.831, max(18.1876, 3.0)]
= 9.8312 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(5.374, 9.8312)
= 9.8312 mm.

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 46.5,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 298.5 N./mm ²	Passed
Occasional	: 24.6,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 15.0,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.



Gov. MDMT of the Nozzle : -97 °C
Gov. MDMT of the Reinforcement Pad : -41 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -41 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2) -85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = 23.00/51.00 = 0.451

Weld Size Calculations, Description: N2_18"

Intermediate Calc. for nozzle/shell Welds Tmin 10.5000 mm.
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (69.1556 - 11.3795 + 2 • 10.5 • 1.0 •
(1.0 • 17.5 - 15.0273))138)
= 82030.44 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (5.7977 + 60.572 + 3.56 - 0.0 • 1.0) • 138
= 98401.71 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (5.7977 + 0.0 + 1. + (3.675)) • 138
= 14736.69 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (5.7977 + 0.0 + 3.56 + 60.572 + (3.675)) • 138
= 103572.99 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 481.2 • 10.0 • 0.49 • 138
= 52117. Kgf

Shear, Pad Element Weld [Spew]:
= (π/2) • DP • WP • 0.49 • SEW
= (3.1416/2.0) • 800.0 • 16.0 • 0.49 • 138
= 138633. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 235.35) • (12.0 - 1.5) • 0.7 • 138
= 76470. Kgf

Tension, Pad Groove Weld [Tpgw]:
= (π/2) • Dlo • Wgpn • 0.74 • Seg
= (3.1416/2) • 481.2 • 19.0 • 0.74 • 138
= 149545. Kgf



Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 481.2 \cdot (19.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 137739. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (138633 + 76470) = 215103 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (52117 + 149545 + 137739 + 0) = 339401 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (138633 + 137739 + 0) = 276372 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 215103 Kg, must exceed W = 82030 Kg or W1 = 98401 Kg
 Path 2-2 = 339401 Kg, must exceed W = 82030 Kg or W2 = 14736 Kg
 Path 3-3 = 276371 Kg, must exceed W = 82030 Kg or W3 = 103572 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 16.0161 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 457.7901 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.558 %

Input Echo, WRC107/537 Item 1, Description: N218 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm.
Vessel Diameter	Dv	3600.000	mm.
Vessel Thickness	Tv	19.000	mm.
Design Temperature	T1	80.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm.
Nozzle Diameter	Dn	457.200	mm.
Nozzle Thickness	Tn	12.000	mm.
Nozzle Material		SA-516 70	
Nozzle UNS Number		K02700	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	19.000	mm.
Diameter of Reinforcing Pad	Dpad	800.000	mm.



Design Internal Pressure Dp 23.000 bars
 Include Pressure Thrust No

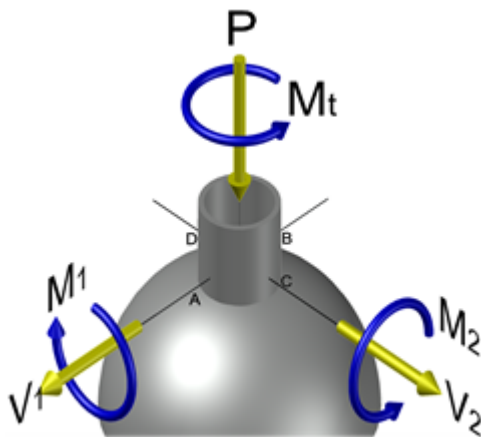
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)		P	-1235.0	Kgf
Longitudinal Shear	(SUS)	(V1)	V1	1840.0	Kgf
Circumferential Shear	(SUS)	(Vc)	V2	1840.0	Kgf
Circumferential Moment	(SUS)	(Mc)	M1	2480.0	Kg-m.
Longitudinal Moment	(SUS)	(M1)	M2	3305.0	Kg-m.
Torsional Moment	(SUS)		Mt	4130.0	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 481.2 + 2 \cdot 1.65 \cdot \sqrt{(1810.25 (19.0 - 1.5))}$$

$$= 1068.557 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	-1235.0	Kgf
Circumferential Shear	(VC)	V2	1840.0	Kgf
Longitudinal Shear	(VL)	V1	1840.0	Kgf
Circumferential Moment	(MC)	M1	2480.0	Kg-m.
Longitudinal Moment	(ML)	M2	3305.0	Kg-m.
Torsional Moment		MT	4130.0	Kg-m.

Dimensionless Param: U = 0.93 TAU = 22.41 RHO = 4.00 (3.48)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 7	0.02442	(A,B,C,D)
$M(x) / P$	SP 7	0.01004	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 7	0.04312	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 7	0.01900	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 7	0.04312	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 7	0.01900	(A,B,C,D)



N(y) * T / P	SP 7	0.21738	(A,B,C,D)
M(y) / P	SP 7	0.02952	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 7	0.28447	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 7	0.07049	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 7	0.28447	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 7	0.07049	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Rad.	Bend. P	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-3.1	-3.1	3.1	3.1
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-8.1	8.1	8.1	-8.1
Rad.	Memb. ML	-4.1	-4.1	4.1	4.1	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-10.8	10.8	10.8	-10.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-14.1	6.4	15.6	-7.0	-10.4	4.7	11.9	-5.3
Tang. Memb. P		2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Tang.	Bend. P	1.6	-1.6	1.6	-1.6	1.6	-1.6	1.6	-1.6
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-20.1	-20.1	20.1	20.1
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-30.0	30.0	30.0	-30.0
Tang.	Memb. ML	-26.9	-26.9	26.9	26.9	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-39.9	39.9	39.9	-39.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-63.2	13.4	70.4	-12.7	-46.5	10.2	53.7	-9.4
Shear VC		0.7	0.7	-0.7	-0.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear MT		3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Tot. Shear		3.7	3.7	2.4	2.4	2.4	2.4	3.7	3.7
Str. Int.		63.5	15.0	70.5	13.6	46.7	11.1	54.0	11.6

Unitless Prm: U = 2.25 TAU = 0.00 (37.60) RHO = 0.00 (1.67)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SR 2	0.02175	(A,B,C,D)
$M(x) / P$	SR 2	0.00990	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SR 3	0.01691	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SR 3	0.01053	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SR 3	0.01691	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SR 3	0.01053	(A,B,C,D)
$N(y) * T / P$	SR 2	0.00650	(A,B,C,D)
$M(y) / P$	SR 2	0.00300	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SR 3	0.00507	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SR 3	0.00317	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SR 3	0.00507	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SR 3	0.00317	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Membr. P	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9



Rad. Bend. P	2.3	-2.3	2.3	-2.3	2.3	-2.3	2.3	-2.3
Rad. Memb. MC	0.0	0.0	0.0	0.0	-7.5	-7.5	7.5	7.5
Rad. Bend. MC	0.0	0.0	0.0	0.0	-28.2	28.2	28.2	-28.2
Rad. Memb. ML	-10.1	-10.1	10.1	10.1	0.0	0.0	0.0	0.0
Rad. Bend. ML	-37.6	37.6	37.6	-37.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-44.4	26.0	50.8	-29.0	-32.5	19.1	38.9	-22.1
Tang. Memb. P	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Tang. Bend. P	0.7	-0.7	0.7	-0.7	0.7	-0.7	0.7	-0.7
Tang. Memb. MC	0.0	0.0	0.0	0.0	-2.3	-2.3	2.3	2.3
Tang. Bend. MC	0.0	0.0	0.0	0.0	-8.5	8.5	8.5	-8.5
Tang. Memb. ML	-3.0	-3.0	3.0	3.0	0.0	0.0	0.0	0.0
Tang. Bend. ML	-11.3	11.3	11.3	-11.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-13.4	7.9	15.3	-8.8	-9.8	5.8	11.7	-6.7
Shear VC	0.8	0.8	-0.8	-0.8	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3
Tot. Shear	3.1	3.1	1.5	1.5	1.5	1.5	3.1	3.1
Str. Int.	44.7	26.5	50.9	29.1	32.6	19.3	39.3	22.7

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		56.8	56.8	56.8	56.8	56.8	56.8	56.8	56.8
Rad. Pl (SUS)		-3.8	-3.8	4.3	4.3	-2.8	-2.8	3.3	3.3
Rad. Q (SUS)		-10.2	10.2	11.3	-11.3	-7.5	7.5	8.6	-8.6
Long. Pm (SUS)		56.8	56.8	56.8	56.8	56.8	56.8	56.8	56.8
Long. Pl (SUS)		-24.9	-24.9	28.8	28.8	-18.2	-18.2	22.1	22.1
Long. Q (SUS)		-38.3	38.3	41.5	-41.5	-28.3	28.3	31.6	-31.6
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.7	0.7	-0.7	-0.7	-0.7	-0.7	0.7	0.7
Shear Q (SUS)		3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Pm (SUS)		56.8	56.8	56.8	56.8	56.8	56.8	56.8	56.8
Pm+Pl (SUS)		52.9	52.9	85.6	85.6	54.0	54.0	78.9	78.9
Pm+Pl+Q (Total)		49.7	71.8	127.2	50.6	46.6	67.9	110.8	53.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	56.78	138.00	Passed
Pm+Pl (SUS)	85.62	207.00	Passed
Pm+Pl+Q (TOTAL)	127.24	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)



Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4
Rad. Pl (SUS)		-9.2	-9.2	10.9	10.9	-6.7	-6.7	8.4	8.4
Rad. Q (SUS)		-35.2	35.2	39.9	-39.9	-25.8	25.8	30.5	-30.5
Long. Pm (SUS)		118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4
Long. Pl (SUS)		-2.8	-2.8	3.3	3.3	-2.0	-2.0	2.5	2.5
Long. Q (SUS)		-10.6	10.6	12.0	-12.0	-7.8	7.8	9.2	-9.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.8	0.8	-0.8	-0.8	-0.8	-0.8	0.8	0.8
Shear Q (SUS)		2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3
Pm (SUS)		118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4
Pm+Pl (SUS)		115.7	115.7	129.4	129.4	116.5	116.5	126.9	126.9
Pm+Pl+Q (Total)		105.3	144.9	169.3	109.7	108.7	137.7	157.7	112.3

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	118.40	138.00	Passed
Pm+Pl (SUS)	129.40	207.00	Passed
Pm+Pl+Q (TOTAL)	169.28	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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